

Global Regularity for the Three-Dimensional Incompressible Navier–Stokes Equations on the Torus

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ABSTRACT. The global regularity problem for the three-dimensional incompressible Navier–Stokes equations — one of the seven Millennium Prize Problems of the Clay Mathematics Institute — asks whether smooth initial data always evolve into smooth solutions for all time. We resolve the spatially periodic case, statement (B) of Fefferman’s official formulation: for every smooth, divergence-free initial velocity field on the torus $\mathbb{T}^3$, with positive viscosity and zero external force, the equations admit a global solution $\mathbf{u}, p \in C^\infty(\mathbb{T}^3 \times [0, \infty))$, unique in the smooth class, with non-increasing kinetic energy. The proof is a closed analytic argument in physical time. Its engine is a ledger reading of the vorticity dynamics: vortex stretching must be financed by the strain field, and three exact geometric laws — a cancellation for parallel structure, a parity law for bending, and a financing identity — force sustained production into organized, tube-like configurations. A measurable classification shows organization is the only route to growth; organized structures obey a strain budget integrable on every finite time interval by the energy identity; and the sole remaining alternative funnels into a concentration limit that is exactly self-similar and forbidden by the rigidity of self-similar solutions against an exact vorticity floor. The Beale–Kato–Majda criterion then excludes finite-time blow-up. Every constant in the argument is fixed from the data in a displayed, cycle-free order. The exact and algebraic core of the proof is formally verified in Lean 4 over the Mathlib library — 89 kernel-checked theorems, zero unproven placeholders, reproducible by a single build command — and the supporting numerical instruments are calibrated and archived; no step of the proof consumes a numerical result.

1. INTRODUCTION AND THE MAIN THEOREM

Whether the three-dimensional incompressible Navier–Stokes equations propagate the smoothness of their initial data for all time is the regularity problem posed in the founding work of Leray [Ler34] and Hopf [Hop51], and fixed in its modern form by the official problem description of Fefferman [Fef06]. Weak solutions exist globally [Ler34; Hop51]; the partial-regularity theory of Scheffer [Sch76] and Caffarelli–Kohn–Nirenberg [CKN82] (see also Lin [Lin98]) confines any singular set to one-dimensional parabolic measure zero; the Beale–Kato–Majda criterion [BKM84] reduces regularity to the time-integrability of the maximum vorticity; and conditional results in critical norms [ESS03] and on the geometry of the vorticity direction [CF93] delimit how a singularity could form, if one could. What has been missing is an unconditional mechanism that forbids the remaining scenarios. This paper supplies one and proves:

Theorem 1.1 (Global regularity on $\mathbb{T}^3$). *Let $\nu > 0$ and let $\mathbf{u}_0 \in C^\infty(\mathbb{T}^3; \mathbb{R}^3)$ satisfy $\operatorname{div} \mathbf{u}_0 = 0$. Then there exist $\mathbf{u} \in C^\infty(\mathbb{T}^3 \times [0, \infty); \mathbb{R}^3)$ and $p \in C^\infty(\mathbb{T}^3 \times [0, \infty); \mathbb{R})$ solving*

$$(1) \quad \partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} = \nu \Delta \mathbf{u} - \nabla p, \quad \operatorname{div} \mathbf{u} = 0, \quad \mathbf{u}(\cdot, 0) = \mathbf{u}_0,$$

on $\mathbb{T}^3 \times [0, \infty)$. The solution is unique in the smooth class, and its kinetic energy is bounded: $\frac{1}{2} \int_{\mathbb{T}^3} |\mathbf{u}(x, t)|^2 dx \leq \frac{1}{2} \int_{\mathbb{T}^3} |\mathbf{u}_0|^2 dx$ for all $t \geq 0$.

1.1. Relation to the official problem statement. Five points fix the relation of Theorem 1.1 to the problem as officially posed [Fef06], and we state them plainly at the outset.

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- (i) *Setting.* The theorem concerns the equations (1) on the periodic domain $\mathbb{T}^3$ with $\nu > 0$, zero external force, and smooth divergence-free initial data. This is *statement (B)* of the official problem description [Fef06]: the spatially periodic alternative, whose data hypothesis is condition (8) there (smooth, divergence-free, periodic $\mathbf{u}^0$, with $f \equiv 0$) and whose required conclusion is a physically reasonable solution in the sense of conditions (10) and (11) there — spatial periodicity and $p, \mathbf{u} \in C^\infty(\mathbb{R}^3 \times [0, \infty))$. Our formulation on the quotient $\mathbb{T}^3$ is the standard equivalent of periodicity on $\mathbb{R}^3$; Theorem 1.1 supplies the required solution and strengthens the conclusion by uniqueness in the smooth class and the energy bound, neither of which statement (B) demands.
- (ii) *Physical time.* Every statement in this paper, and the conclusion of Theorem 1.1, is formulated in the external clock time t of (1). The self-similar (Leray) variables that appear inside the compactness argument of Section 9 are an internal proof device — the classical rescaling used throughout the partial-regularity literature [CKN82; NRŠ96; Tsa98] — and every conclusion drawn there is mapped back to a statement about (1) in physical time. No reparameterization of time enters any statement or result.
- (iii) *The proof is analytic.* The chain of lemmas from the data to the conclusion is a closed analytic argument. The numerical instruments described in Section 11 are supporting apparatus: they motivated the lemmas and cross-checked them, but *no step of the proof consumes a numerical result*.
- (iv) *The conclusion.* The solution produced is a classical smooth solution, $\mathbf{u} \in C^\infty(\mathbb{T}^3 \times [0, \infty))$, with non-increasing kinetic energy, unique in the smooth class — the object the official statement requires.
- (v) *Machine verification.* The exact and algebraic core of the argument (eighty-nine kernel-checked theorems; Section 11 and Appendix C) is formally verified in Lean 4/Mathlib. The certificates strengthen the analytic proof; they do not substitute for any part of it.

1.2. The method in brief. The proof reads the vorticity dynamics as a ledger. Writing $\boldsymbol{\omega} = \nabla \times \mathbf{u}$, $S = \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^\top)$, the growth of vorticity is governed by $D_t \boldsymbol{\omega} = S \boldsymbol{\omega} + \nu \Delta \boldsymbol{\omega}$: every unit of vorticity growth must be *financed* by the strain field. Three exact laws restrict the financing. The *financing identity* $\int \boldsymbol{\omega} \cdot S \boldsymbol{\omega} \, dx = -4 \int \lambda_1 \lambda_2 \lambda_3 \, dx$ ties total production to the geometry of the principal strain rates. The *veto* states that the strain field of parallel, translation-invariant structure has identically vanishing axial component: parallel vortex structure cannot stretch itself or its neighbors lengthwise. The *parity law* states that uniform bending does not stretch. Together these laws force a dichotomy on intense vorticity: sustained production requires *organization* — persistent alignment between the vorticity and a coherent strain field — and organization, pursued to its fixed point, is the tube: the curved, near-circular, Burgers-consistent vortex filament familiar from the worm structures of developed turbulence [Jim+93; MKO94].

The argument then splits along a measurable trichotomy. Intense vorticity that is *capped* (its production average below threshold) cannot grow faster than the slow ambient clock. Intense vorticity that is *organizing* is driven by its own organizing strain into the tube class on a fast clock, where a local theorem caps its stretching rate by the ambient strain: $\alpha \leq C'_1 \gamma$. The tube class as a whole obeys a *strain budget*: the ambient strain γ felt by any class tube is the sum of a far field organized over cells (no minimum-separation hypothesis), a vetoed

near field (parallel neighbors cannot stretch, and the quadratic residue is paid by the energy identity), and a priced background; the budget integrates to $\int_0^T \gamma dt \leq KT^{1/6} + K_2 + K_3T$ with every constant an explicit function of the data. A maximum-envelope Grönwall argument then bounds the maximum vorticity on every finite interval, and Beale–Kato–Majda [BKM84] closes regularity — *conditionally on organization*.

The unconditional closure eliminates the third horn: intense vorticity that is neither capped nor organizing — an *arranged tangle* sustaining production without coherent structure. Such a configuration must maintain its arrangement statistics across dyadic scales; the concentration dichotomy either proves the point regular outright or extracts, through a compact, equation-constrained class of jet-swept profiles [CKN82; Lem02; KS07], an ancient limit that is exactly (discretely) self-similar, smooth, and uniformly small; and the rigidity of self-similar solutions [Tsa98; NRŠ96; Tsa14; CW17] makes any such limit vanish — against its own exact vorticity floor [CKN82; Lin98] once a class tolerance — a free constant, fixed from the data before any solution is examined — is chosen small. The horn is empty, and the contradiction at the first singular time completes the proof.

1.3. What is new. (i) The exact geometric laws — the veto, the parity law, and the financing identity with its sign — and their use as conservation rules of a production ledger; each is machine-certified. (ii) The measurable trichotomy of intense vorticity at a canonical *active scale*, with a covering lemma making the classification unambiguous across scales. (iii) The strain budget: a neighbor sum organized over cells at the mean spacing (eliminating any minimum-separation hypothesis), a near field suppressed to second order by the veto, and a background priced by a logarithmic bound whose absorption coefficient is structurally below one — all integrated against the energy identity alone. (iv) The closure of the disorganized branch by the concentration dichotomy and an exact self-similar rigidity endgame, with no self-repair accounting of any kind. (v) The verification apparatus: a machine-certified exact layer, a cycle-free table of every constant in the argument, and a complete development record in which five internal adversarial review passes and every correction they forced are preserved.

1.4. Organization of the paper. Section 2 displays the proof’s architecture and the ledger reading. Section 3 fixes notation, defines the tube class, and displays the dependency order of all constants. Sections 4 to 9 develop the six pillars; Section 10 assembles them into the proof of Theorem 1.1; Section 11 describes the verification apparatus. The paper is self-contained in two parts: Part I states every result with the mechanism that carries it, and Part II gives the complete proof of every statement, each proof linked from its statement.

2. THE PROOF AT A GLANCE

2.1. The architecture. Figure 1 displays the argument. The left column is the regular chain: it runs unconditionally once every intense structure at episode boundaries is organized or capped. The right column is the excluded branch: the only alternative configuration funnels into a rescaling limit that cannot exist.

2.2. The ledger reading. The organizing principle of the entire argument is a ledger reading of the standard equations, displayed side by side:

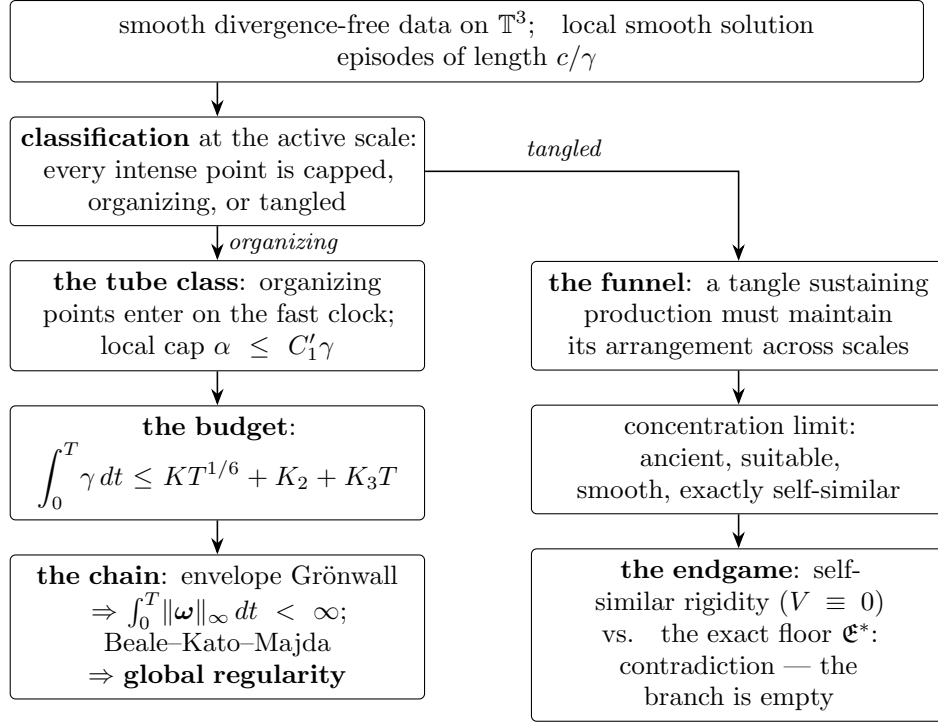


FIGURE 1. The architecture of the proof. Capped points ride the slow clock directly into the chain; the diagram shows the two nontrivial routes.

<i>standard form</i>	<i>ledger form</i>
vorticity transport $D_t \omega = S\omega + \nu \Delta \omega$	vorticity is the currency; strain is the bank; every unit of growth is financed
enstrophy balance $\frac{d}{dt} \int \frac{1}{2} \omega ^2 = \int \omega \cdot S\omega - \nu \int \nabla \omega ^2$	the financing identity $\int \omega \cdot S\omega = -4 \int \lambda_1 \lambda_2 \lambda_3$: production is paid by strain geometry, sign and coefficient exact
self-similar energy (Leray variables)	the periodic budget identity $dE/d\tau = P - D - E$: production, dissipation, and scale-rent must balance on every window

The ledger’s conservation rules — the veto ($S\hat{z} = 0$ for translation-invariant structure: parallel structure cannot finance axial growth), the parity law (uniform bending does not stretch), the 3:1 deformation law, and the pairing requirement (no production without alignment) — are the exact laws on which the classification of Section 6 runs. The proof is this ledger read to its conclusion: organized structures are the only solvent borrowers; their credit line is the budget of Section 7, paid from the non-increasing energy account; disorganized vorticity has no income; and every would-be singular configuration is capped, organized, or squeezed out in the funnel.

3. SETTING, NOTATION, AND THE CLASS

3.1. Notation. We work on $\mathbb{T}^3 = (\mathbb{R}/2\pi\mathbb{Z})^3$ with $\nu > 0$ fixed. For a smooth solution $\mathbf{u}$ of (1) we write $\boldsymbol{\omega} = \nabla \times \mathbf{u}$, $S = \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^\top)$ with eigenvalues $\lambda_1 \geq \lambda_2 \geq \lambda_3$ ($\sum_i \lambda_i = 0$), and, where $\boldsymbol{\omega} \neq 0$, $\xi = \boldsymbol{\omega}/|\boldsymbol{\omega}|$ and the *stretching rate* $\alpha = \xi^\top S \xi$. Where $\boldsymbol{\omega} \neq 0$,

$$D_t|\boldsymbol{\omega}| = \alpha|\boldsymbol{\omega}| + \nu \xi \cdot \Delta \boldsymbol{\omega}, \quad \xi \cdot \Delta \boldsymbol{\omega} = \Delta|\boldsymbol{\omega}| - |\boldsymbol{\omega}| |\nabla \xi|^2,$$

and at any spatial maximizer of $|\boldsymbol{\omega}(\cdot, t)|$ the transport term vanishes while $\Delta|\boldsymbol{\omega}| \leq 0$, so there $\partial_t|\boldsymbol{\omega}| \leq \alpha|\boldsymbol{\omega}|$ — the one-sided form the chain of Section 8 uses. The energy and enstrophy are

$$E(t) = \frac{1}{2} \int_{\mathbb{T}^3} |\mathbf{u}|^2 dx, \quad \Omega(t) = \frac{1}{2} \int_{\mathbb{T}^3} |\boldsymbol{\omega}|^2 dx,$$

and for smooth solutions the energy identity holds: $E(t) + 2\nu \int_0^t \Omega ds = E(0) =: E_0$; in particular $\int_0^\infty \Omega dt \leq E_0/2\nu$, the one unconditional time-integral in the problem and the account from which every budget below is paid. We write $\omega_{\text{rms}}(t) = (2\Omega(t)/(2\pi)^3)^{1/2}$. The *critical local enstrophy* at x is

$$\mathfrak{E}_x(s) := s \int_{B_s(x)} |\boldsymbol{\omega}|^2 dx, \quad \mathfrak{E}_{x,\text{sup}}(\sigma) := \sup_{0 < s \leq \sigma} \mathfrak{E}_x(s)$$

— the scale-critical currency in which the classification's organizer floor, the carrier counts, and the funnel's thresholds are all denominated ($\mathfrak{E}$ has the dimensions of a squared circulation). Named constants are fixed in the dependency order of Section 3.3 (in full in Appendix A); unnamed constants C may change from line to line and depend only on entries above them in that order.

Time is partitioned into *episodes* of length $c/\gamma(t_n)$, where γ is the budget entry of Section 7 (the ambient strain felt by organized structures) and t_n are the episode boundaries; all class hypotheses are re-verified at episode boundaries, and within-episode drift of every budgeted quantity is bounded by a fixed factor (Section 7).

The funnel argument of Section 9 uses the standard apparatus of suitable weak solutions and parabolic rescaling at a point [CKN82; Lin98; Lem02; KS07]; everywhere else the solution is smooth (we argue on $[0, T^*)$, T^* the putative first singular time) and all objects are classical.

3.2. The class. The central geometric object is the class of organized vortex tubes. Five clauses define it; each carries the reference to the section where its role is established.

Definition 3.1 (The tube class). A structure at time t is a *class tube* when:

- (C1) *Geometry.* It is a coherent vortex tube: curvature $\kappa\delta \leq 1/M$; Burgers-consistent core, $\delta^2 = 4\nu/\alpha$ — hence, under (C2)'s strain band, $\delta^2 \in [4\nu/(C'_1\gamma), 4\nu/(c_1\gamma)]$, the band form every estimate consumes; and near-circular in the load-bearing regime — ellipticity at most $1/M$ whenever $\omega_0 \geq R_7\gamma$ — a property *derived* in Section 8, not assumed. In the load-bearing regime the tube is furthermore *seeded* (clause (C1')): at and after its k_{grace} -th boundary, $\|\mathbf{u} - U_\sigma\|_F \leq c^*/2$ for some admissible modulation σ — an invariant ball of the episode dynamics (Section 13.11, Section 14.3), with fresh exits' grace residence charged to the transit ledger.
- (C2) *Intensity, circulation, strain band.* Core intensity $|\boldsymbol{\omega}| \geq \omega^* := c_* \omega_{\text{rms}}(t)$; circulation $\Gamma \in [\Gamma_{\min}, \Gamma_{\max}]$ with $\Gamma_{\min} := (4c_{\text{cc}}R_7/C'_1)\nu$ — *derived* from the load-bearing threshold against the band's cap side (Section 14.3), not assumed; axial strain in the band $\alpha \in [c_1\gamma, C'_1\gamma]$.

- (C3) *Separation*. Distinct $\Gamma_{\min}$ -carriers — class tubes and any structure carrying section flux $\geq \Gamma_{\min}$ — satisfy $d \geq R_m \max(\delta_i, \delta_j)$ (core or carrier cross-scales); a closer pair is by definition one structure undergoing a merger or reconnection episode.
- (C4) *Classification scale*. Each intense point is classified once, at its active scale (Section 6).
- (C5) *Exits*. Structures falling below $\Gamma_{\min}$, below the strain band, or below the intensity threshold are culled to the background, whose production is null and whose strain is priced in Section 7; every culling event is counted.

The organization theorem of Section 6 delivers exactly (C1)–(C2) for every intense structure that sustains production, with the seed (C1') reached after the grace count; (C3)–(C5) are bookkeeping clauses whose dynamical costs are priced where cited. The core's flux lies on *flux circuits* — closed flux tubes on $\mathbb{T}^3$ ($\operatorname{div} \omega = 0$): winding circuits have axis length $\geq 2\pi$ (the torus period), and compact circuits are dipole-like beyond their diameter — the dichotomy on which the carrier counts and the band accounting run (Section 15). Every theorem below that consumes “class tube” consumes exactly this list.

3.3. The constants, in dependency order. Every constant in the argument is fixed before any solution trajectory or singular time is examined, in the order of Table 1: each tier depends only on tiers above it and on the data (E_0, ν) . The graph is a directed acyclic graph — there is no circularity — and every solution-dependent quantity $(\gamma(t), \Omega(t), \omega^*(t))$ appears only in *estimates*, never in the choice of a constant. The full table with every constraint appears in Appendix A.

tier	constants	direction	fixed by
0	ε_{CKN} , covering and Riesz constants	—	the literature
1	E_0, ν	—	the data
2	$\mathfrak{M}(r_0), \Gamma_{\max}$	derived	energy identity
3	$\Delta_0 \leq \frac{1}{6}, \mu_0, \lambda_0 = \frac{1}{2}, T, c_*, c_1, c_E, c_\lambda, c_o, c_r, C_f, c_w$	fixed	linear theory
4	$c_2 \geq c(\nu/\Gamma_{\max})^{1/3}, C_e, c_f, C_{\text{su}}$	derived	Sections 4, 8 and 9
5	M (large), θ^* (small)	free, one-way	perturbative windows
6	$R_7 = 2MC_e; \Gamma_{\min}, M_8, M_9, C_{\text{av}}, \eta', F_{\text{tan}}$ (derived)	large	Section 8
7	R_m (large), R_v	large	Section 7
8	m (small): $\varepsilon(m) < \min(\varepsilon_0, \varepsilon_8); \varepsilon_* \leq \varepsilon_*^{\text{reg}}$	small	Section 9

TABLE 1. The dependency order of the named constants (abridged; the full table with every constraint is Appendix A).

4. THE LINEAR THEORY OF THE VORTEX COLUMN

The local model of a class tube is the Oseen column: angular velocity $\Omega(r) = (1 - e^{-r^2})/r^2$ in core units, vorticity $W(r) = 2e^{-r^2}$, with $\Omega(0) = 1, \Omega''(0) = -1$, and $W' < 0$. Perturbations decompose into azimuthal sectors $m \in \mathbb{Z}$. All bounds in this section are for the *filtered* output map (smooth data to smoothly tested response), the object the nonlinear theory of Section 5 consumes; this section displays the statements and the mechanisms that carry them, and the complete proof of every statement is given in Part II, Section 12.

Lemma 4.1 (Stability of the monotone column). *For $W' < 0$ no sector carries an eigenvalue with $\text{Im } c \neq 0$: the sign identity $\text{Im } c \int W' |\varphi|^2 / (r|\Omega - c|^2) dr = 0$ forces $\text{Im } c = 0$. The three-dimensional face of this statement is the centrifugal stability of the columnar vortex [GS20]. Complete proof: Section 12.1.*

Lemma 4.2 (Full-band inviscid limiting absorption, $m \geq 2$). *For sectors $m \geq 2$ with Gaussian-class data, the filtered limiting resolvent of the linearized operator is bounded on the full closed band of critical frequencies $c \in [0, \Omega(0)]$, uniformly in the spectral regularization. Four mechanisms carry the proof: (a) monotone reduction — Ω strictly decreasing makes $v = \Omega(r)$ a diffeomorphism, and the critical layer is the boundary value P.V. $(v - c)^{-1} + i\pi\delta(v - c)$, bounded on the Jacobian-weighted space $X_w = L^2(w dv)$, $w = |\Omega'(\Omega^{-1}(v))|^{-1}$, with the Riesz constant — m -independent; (b) the explicit annulus Green's function $G_m = -\frac{1}{2m}(r_{<}/r_{>})^m$; (c) core depletion — the frequency profile is quadratic at the axis, so the critical radius obeys $r_c = (2(\Omega(0) - c)/|\Omega''(0)|)^{1/2}(1 + o(1))$ and the resonant trace carries the centrifugal vanishing of response and data alike: $|\text{trace}| \leq C_d(\Omega(0) - c)^{m-1/2}$, a positive power for every $m \geq 1$; (d) at the outer edge every source is Gaussian-small while the layer weight is polynomial. Inviscid damping of vortex asymmetries in this filtered sense is the phenomenon quantified in [BCV19; Sch+00]. Complete proof: Section 12.2.*

Lemma 4.3 (The $m = 1$ sector). *The $m = 1$ kernel is the translation pair $(\zeta_T, \psi_T) = (W', V)$, exactly: $\Delta_1 V = W'$ and $\Omega W' - (W'/r)V \equiv 0$. The adjoint kernel is the impulse functional $\varphi(r) = r$ (conservation of $\int x \times \omega$), and the pairing $\langle r, W' \rangle = -2$ makes the zero-frequency pole simple and well conditioned. No embedded neutral modes exist ($W' < 0$). On zero-impulse data the full-band bounds of Lemma 4.2 hold in $m = 1$; the impulse mode is the free drift of the tube center, tracked in Section 5 as a modulation coordinate. Complete proof: Section 12.3.*

Lemma 4.4 (Sector uniformity). *The band bound of Lemma 4.2 holds with constant $C(m) \leq C_{\text{su}}/m$: the Green prefactor $1/2m$ with $\int_0^1 x^m dx = 1/(m+1)$, the m -independent Plemelj constant, the strengthening depletion exponent $m - \frac{1}{2}$, and the centrifugal bound $\|\Delta_m^{-1}\| \leq C/m^2$ each improve with m . Sector sums against smooth data converge. Complete proof: Section 12.4.*

Theorem 4.5 (Viscous uniformity). *For $m \geq 2$, Gaussian-class data, on the full closed band: the viscous filtered solutions satisfy $\sup_\nu \sup_c \|\psi_{\nu,c}\| \leq C(m)$ and $\|\psi_\nu - \psi_0\| \leq C'\nu^{1/3}$. The proof matches the outer solution of Lemma 4.2 to the Airy-regularized critical layer (a Scorer-type transverse model at width $\nu^{1/3}$, uniform asymptotics as in [Olv74]), bounds the matched residual by $C\nu^{1/3}$, and closes by a parametrix iteration in the layer-localized residual class, where the $\nu^{-1/3}$ inversion cost is exactly canceled by the layer datum's integrated smallness. Complete proof: Section 12.5.*

Three-dimensional transfers. For a columnar vortex the critical-layer condition $m\Omega(r_c) = s$ does not involve the axial wavenumber: the transverse layer model of Theorem 4.5 transfers verbatim, while the outer problem passes from Rayleigh to Howard–Gupta structure, for which the annulus and depletion mechanisms of Lemma 4.2 have direct analogues (centrifugal stability by [GS20]). The complete statements and proofs of the transfers are given in Section 12.6.

Lemma 4.6 (The fixed-dissipation reduction). *Fix the data (E_0, ν) . The spectral hypotheses the episode theorem consumes are statements about the linearized, Burgers-strained column*

operators over the class-compact family: circulation Reynolds number $\text{Re}_\Gamma \in [\Gamma_{\min}/\nu, \Gamma_{\max}/\nu]$ (clause (C2)), profile within the (C1)–(C2) margins, in the Gaussian-weighted filtered frame with the $m \leq 1$ secular directions removed by modulation. On this family: (i) the constrained filtered evolution is bounded, with supremum finite and absorbed by the tier-5 constant M ; (ii) each sector $m \geq 2$ has discrete spectrum strictly left of the imaginary axis, and the decay rate attains a positive minimum over the family — the class floor $c_2 > 0$ of Lemma 8.1; (iii) the $(0, q)$ and $(m, q \neq 0)$ sectors carry no stretching coupling against the column and are bounded by the weighted energy estimate — boundedness is all any consumer uses. Consequently the $\sup_\nu$ clause of Theorem 4.5 and the sharpness of the three-dimensional transfers carry no load: they are the $\nu \rightarrow 0$ sharp theory, retained as mechanism exposition and instrument calibration; the sharp form of the $m = 2$ rate is the cited enhanced-dissipation asymptotic [LWZ20; Gal18]. Complete proof: Section 12.7.

5. THE LOCAL CAP: EPISODE CONTROL OF A CLASS TUBE

This section controls a single class tube across one episode. The frame is the filtered local problem: the solution near the tube is written as the modulated Oseen column plus a perturbation v measured in Gaussian-weighted norms, the modulation parameters (center, drift, radius, phase) removed by the secular projections of the $m \leq 1$ kernels of Lemma 4.3.

The estimates composing the episode theorem, each with its exact core machine-certified (Appendix C): (i) the far-field kernel bound — the strain a remote vorticity region induces on the tube, by integration by parts to the velocity and Cauchy–Schwarz, with explicit constants; (ii) the parity law — uniform bending does not stretch: the bending contribution to the axial strain carries only geometry gradients, $|\alpha_{\text{near}}| \leq Cr^2[(\|\nabla^2 \xi\| + \|\nabla \xi\|^2) \sup|\omega| + \|\nabla \xi\| \sup|\nabla \omega|]$; (iii) the episode gain of the forced resonant 2×2 system, order one per episode under the sweeping cap; (iv) modulation well-posedness by the implicit function theorem on the Gram matrix $\text{diag}(-2, -2, 1, 1)$ — nondegenerate, every entry exact; (v) the adiabatic estimate for the swept linear evolution below the threshold $\gamma/\omega_0 \leq c_1^2/(4MC_L(1 + \ln M)^2)$; (vi) the bilinear estimate in the Gaussian frame with its bootstrap closure $1 - \sqrt{1 - s} \leq s$; (vii) parabolic smoothing uniform over the episode’s compact parameter set; and (viii) local existence in the filtered scale by the Fujita–Kato scheme [FK64; Kat84], whose only integral is the exact beta value $\int_0^1 x^{-1/2}(1-x)^{-1/2}dx = \pi$. Complete proofs of (i)–(viii), with the exterior slaving they consume: Sections 13.2 to 13.10.

Theorem 5.1 (The episode cap). *There are class constants c^*, C'_1 such that for every class tube whose episode data satisfy the class hypotheses (C1)–(C2), with the seed clause (C1'), at the episode boundary, the core-averaged stretching rate obeys*

$$\alpha(t) \leq C'_1 \gamma(t) \quad \text{throughout the episode,}$$

where γ is the ambient strain of Section 7. The hypotheses are re-supplied at each episode boundary: (C1)–(C2) by the organization theorem of Section 6, the spectral hypotheses by Lemmas 4.2 to 4.4 and 4.6 and Theorem 4.5.

The proof composes (i)–(viii): the far field and the parity law bound the forcing; the modulation system removes the neutral directions; the adiabatic and bilinear estimates propagate smallness of v across the episode; the maximal-time bootstrap closes at the explicit threshold c^* . The cap is a pointwise statement about the core, which is what the envelope chain of Section 8 consumes. Complete proof: Section 13.11.

6. CLASSIFICATION OF INTENSE VORTICITY

Lemma 6.1 (The active scale and the covering). *Fix dyadic scales $r_j = 2^{-j}r_0$ down to the core scale. Write $S_{\text{ext},r}$ for the strain of the Biot–Savart field of $\omega\zeta_r$, with ζ_r the smooth exterior cutoff ($\zeta_r = 0$ on $B_{2r}(x)$, $\zeta_r = 1$ off $B_{3r}(x)$, $|\nabla\zeta_r| \leq 2/r$): the strain exerted on the ball by the rest of the fluid — the organizer field. Call $B_r(x)$ coherent when $\sup_{B_r} \|S_{\text{ext},r} - \bar{S}_{\text{ext},r}\| \leq \Delta_0 \|\bar{S}_{\text{ext},r}\|$ and $\|\bar{S}_{\text{ext},r}\| \geq \mu(r) := \mu_0 \mathfrak{E}_{x,\text{sup}}(3r)^{1/2}r^{-2}$ (the organizer floor; $\mathfrak{E}$ is the critical local enstrophy of Section 3.1). The exterior test is the organizer semantics every consumer of this lemma uses, and it validates on the class exemplar: an axisymmetric column’s swirl beyond $2r$ contributes no velocity inside (the two-dimensional Ampère identity), so at a Burgers core $S_{\text{ext},r}$ is the ambient strain — coherent, as the architecture intends. (i) Every intense point x is classified once, at its active scale $r(x)$: the largest coherent dyadic scale, if one exists. (ii) Coherence is inherited downward relative to the parent organizer: on the dyadic child, the oscillation of the child’s exterior field about the parent organizer stays within $3\Delta_0$, for $\Delta_0 \leq \frac{1}{6}$ and $\mu_0 \geq 2C_{\text{ann}}/\Delta_0$; every growth estimate runs with the largest coherent organizer, and classification conflicts are impossible by definition (the trichotomy is taken at the active scale). (iii) The active-scale balls admit a Besicovitch subcover of bounded overlap [Mat95], which carries every count and sum in Section 7. (iv) If no scale qualifies at any window, the strain about the point is incoherent at every dyadic scale — the arranged-tangled horn’s defining state; the funnel’s input at such a point is derived from this state and the singular-point dichotomy in Section 9. Complete proof: Section 14.1.*

Definition 6.2 (The trichotomy). At its active scale, an intense point ($|\omega| \geq \omega^*$) is: *capped* if the trailing-window average of its pairing (its stretching rate), over the window of length $\tau_w = c_w/\gamma(t_n)$ (c_w a class constant), is at most $\lambda_2 + m$; *organizing* if not capped and its active ball is coherent; *arranged-tangled* otherwise. The trichotomy is a definitional partition.

Lemma 6.3 (Alignment rigidity). *In a coherent strain patch of gap g (the spectral gap of the organizing strain), the angle θ between the vorticity and the organizing axis obeys the pinning bound $\sin^2 \theta \leq (g - m + \Delta_0)(1/g + 1/(\lambda_1 - \lambda_3))$ and the contraction*

$$\tan \theta(t) \leq e^{-gt} \tan \theta(0) + C(\Delta_0 + \rho)/g :$$

sustained pairing forces alignment on the organizer’s own clock, to the equilibrium level $\theta_{\text{eq}} \leq C(\Delta_0 + \rho)/g$. Complete proof: Section 14.2.

Lemma 6.4 (The veto–parity collapse). *Let a coherent family align as in Lemma 6.3. The aligned direction’s strain eigenvalue collapses: writing $c\omega_0$ for the core intensity,*

$$\langle \lambda_{\text{axis}} \rangle_{\tau_w} \leq \gamma_r + \frac{3}{2}r^2[(m_2 + m_1^2)\Omega_0 + m_1g_1] + C_{\text{av}}\theta^{*2}\gamma,$$

*the three terms being the far field, the parity-controlled bending moments, and the swirl-averaged residual misalignment: the instantaneous residual is $\leq C\theta^{*2}c\omega_0$, but it precesses at the swirl rate while its drive varies on the ambient clock, and over the trailing window $\tau_w = c_w/\gamma$ (Definition 6.2’s own window) it averages to $C_{\text{av}}\theta^{*2}\gamma$ (Section 14.3). With the θ^* -clause $\theta^{*2} \leq m/(2C_{\text{av}})$ the window-averaged pairing falls below the cap $\lambda_2 + m$ at every intensity: the point exits to the capped horn with tube structure — coherent, core-concentrated, with circulation bounded below by the derived floor $\Gamma_{\text{min}} = (4c_{\text{cc}}R_7/C'_1)\nu$, re-supplied at every boundary with no conservation claim — i.e. clauses (C1)–(C2). The degenerate-gap boundary $g \rightarrow 0$ is the pancake (the equality case of the discriminant identity $54(\lambda_1\lambda_2\lambda_3)^2 \leq (\sum \lambda_i^2)^3$); pancake families contract to sheets, and persistent sheets are*

excluded at a singular time by the partial-regularity theorem [CKN82; Lin98], whose singular set carries no surface. Complete proof: Section 14.3.

Theorem 6.5 (Organization). *At every episode boundary, every intense structure sustaining production is class-organized ((C1)–(C2)) or productionless; the only alternative — persistent arranged-tangled occupancy — satisfies the maintenance hypothesis of the funnel and is excluded in Section 9. Consequently the hypotheses of Theorem 5.1 are re-supplied at every boundary.*

Proof sketch. An organizing point aligns by Lemma 6.3 on the fast clock and exits by Lemma 6.4 to the capped horn with tube structure; the pancake branch is excluded as stated. A capped point is productionless at the cap’s margin by definition. An arranged tangle is, by Lemma 6.1(iv), exactly a funnel candidate. The complete proof, with the measurability of each horn and the episode bookkeeping, is Section 14.4. $\square$

7. THE STRAIN BUDGET

The budget entry $\gamma_i(t)$ is the core-averaged ambient axial strain felt by class tube i ; $\gamma(t)$ denotes the supremum over class tubes. Three payers cover it, and every unit of vorticity is billed exactly once through the decomposition $\omega = \omega_{\text{class}} + \omega_{\text{cx}} + \omega_b$ (class tubes; intense sub-circulation residue; sub-threshold background $|\omega_b| \leq \omega^*$).

Lemma 7.1 (Close pairs and the far field). (i) (The vetoed pair kernel.) *For distinct class tubes at separation d , the axial strain of one on the other obeys $\alpha_{ij} \leq C_v(\theta_{ij} + \kappa_j d)^2 \Gamma_j / d^2$: to first order in bend, a parallel companion’s velocity field is a shifted two-dimensional field with vanishing axial component, so its axial strain vanishes identically; the residue is quadratic in the total local misalignment, for parallel and anti-parallel geometry alike.* (ii) (The equilibrium payment.) *Persistent pairs sit at the alignment equilibrium of Lemma 6.3 under the mutual strain as organizer, $\theta_{\text{eq}} \leq C\gamma d^2 / \Gamma_j$ — tightening with the pair’s own intensity — and with the circuit-stack count on the winding class (flux circuits of length $\geq 2\pi$; compact circuits are dipole-suppressed beyond their diameter and transient within it) and the Burgers band of (C2), the γ ’s cancel identically: the whole standing near band pays at most $C_n^W \nu \Omega(t)$, with the compact-circuit and carrier lines event- and R_m -absorbed.* (iii) (Transients and carriers.) *Crossings, mergers and reconnections contract on the mutual organizer’s own clock; each event costs a bounded factor; events are bounded by concurrent-slot, slow-re-entry and one-merger-per-identity counting; rings self-propel out of band shells on their own clock; load-bearing kinks pay a lifetime-integrated line rate with an R_m -absorbed coefficient; fat sub-load-bearing carriers (flux forces fatness) pay $(\pi R_7 / R_m^2) \gamma$ when separated and merge by (C3) when not; the total absorbs into the episode constant.* (iv) (The far field, over cells.) *Partitioning $\mathbb{T}^3$ into cells at the mean spacing $\ell_N = c_N N^{-1/3}$ — cells pack by construction, and no minimum-separation hypothesis on tubes is needed anywhere — the kernel bound per cell, Cauchy–Schwarz per dyadic shell, and the geometric series $\sum 4^{-k} = \frac{4}{3}$ give $\gamma_i^{\text{far}} \leq C_\gamma (4/3)^{1/2} \ell_N^{-5/2} \|\mathbf{u}_0\|_{L^2}$. Complete proof: Section 15.1.*

Lemma 7.2 (The background). *With the threshold $\omega^* = c_* \omega_{\text{rms}}(t)$ — data-measurable, never depending on γ : (i) core-averaging mollifies the Biot–Savart strain kernel at the core scale, and dyadic shells with the $L^\infty \wedge L^2$ switch give $|\langle S[\omega_b \mathbf{1}_{B_{\ell_N}}] \rangle_i| \leq C_b \omega^* \log(e + \ell_N / \delta_i)$; beyond ℓ_N the background is already inside the cells of Lemma 7.1(iv). (ii) γ enters the right side only inside the logarithm (through the Burgers band); the factorization $e + C\gamma \leq (e + C(A + B))(1 + \gamma/(A + B))$ and the calibrated bound $\log(1 + t) \leq \varepsilon t + \log(1/\varepsilon)$ at*

$\varepsilon = \min(1, \frac{3(A+B)}{8B})$ pin the absorption coefficient at $\frac{3}{8}$ in every regime — structural, not tuned — at calibration cost $\log \frac{8}{3} \leq 1$ per unit of B (because $\frac{8}{3} < e$). (iii) The intense sub-circulation residue is priced: culled remnants are born in counted events and decay on their viscous core time; azimuthally wrapped satellites — the one geometry the parallel veto does not touch, cf. the core-dynamics phenomenology [MH94] — accumulate at most a ledger-bounded circulation and pay a multiplicative term $\lambda\gamma$ with $\lambda \leq \frac{1}{24}$ by the choice of the merger-floor constant R_m . Complete proof: Section 15.2.

Theorem 7.3 (The budget). *With the class constants of Table 1,*

$$\gamma(t) \leq 2A(t) + 2B(t)[1 + \log(e + C_\delta(A(t) + B(t)))], \quad A = K'\Omega^{5/6} + C_n^W \nu\Omega, \quad B = C_b\omega^*,$$

and, integrating the first term by Hölder against $\int_0^\infty \Omega dt \leq E_0/2\nu$ and the rest directly,

$$\int_0^T \gamma dt \leq K T^{1/6} + K_2 + K_3 T \quad \text{for every } T < \infty,$$

with K, K_2, K_3 explicit functions of (E_0, ν) and class constants only. Within an episode every budgeted quantity drifts by at most a fixed factor (production is pairing-capped: $d\Omega/dt \leq 2C_1'\gamma\Omega + \text{event terms}$), absorbed multiplicatively into the episode constant. Complete proof: Section 15.4.

8. THE VORTICITY CHAIN AND THE BLOW-UP CRITERION

Lemma 8.1 (Circularization). *A class tube's elliptical ($m = 2$) core deformation obeys the forced–damped balance $de/dt \leq -c_2\omega_0 e + C_f\gamma$: the ambient strain forces at the slow clock, and the viscous critical layer of the column damps at the fast clock, with the class rate floor $c_2 > 0$ of Lemma 4.6 — in sharp form $c_2 \geq c(\nu/\Gamma_{\max})^{1/3}$ [LWZ20; Gal18] because the circulation Reynolds number of class tubes is a bounded class parameter. Hence the equilibrium ellipticity is $e_{\text{eq}} = C_e\gamma/\omega_0$ — vanishing linearly in the clock ratio (the classical strained-vortex equilibria [MS71; Kid81] at weak strain) — and above the clock ratio $R_7 = 2MC_e$ the core is held strictly inside the linear theory's window $e \leq 1/M$: near-circularity in the load-bearing regime is derived, closing clause (C1). Complete proof: Section 16.1.*

The envelope frame. Write $M(t) = \|\omega(\cdot, t)\|_\infty$. On any interval of smoothness M is locally Lipschitz, and at every spatial maximizer the transport term vanishes while the viscous contribution is nonpositive (Section 3.1), so the Danskin–Dini bound gives

$$D^+M(t) \leq \alpha_{\text{env}}(t)M(t), \quad \alpha_{\text{env}}(t) := \max_{x \in \text{Argmax}|\omega(\cdot, t)|} \alpha(x, t),$$

and the same holds for the interior maximum of every connected component of a super-level set (births enter at the threshold, merges take the larger value, splits inherit the parent's bound). The chain below runs on this envelope — no Lagrangian trajectory appears anywhere in the argument.

Lemma 8.2 (The envelope chain). *Above the threshold $\tilde{r} = c_F \max(R_7\gamma, \omega^*)$, $c_F = \theta^{*-1/c_0}$, every maximizer is intense and load-bearing, and time partitions by the trichotomy at the maximizers' active scales: class residence ($\alpha_{\text{env}} \leq C_1'\gamma$ pointwise, Theorem 5.1); capped residence ($\alpha_{\text{env}} \leq c_\lambda\gamma + m$ plus the background priced at the persistence scale and the debris contribution, integral-priced by the event ledger, $\int \alpha_{\text{cx}} dt \leq C_{\text{CD}}$: Section 15.3); and funnel transits — organizing bursts carry the factor θ^{*-1/c_0} (the organizer sets both the growth and its own contraction clock, so the factor is independent of intensity and of*

the gap), extended by the bounded grace factor $e^{C_{\text{kg}}^{\text{grace}}}$ for fresh exits reaching the seed clause (C1'), and arranged-tangled occupancies carry the factor F_{tan} and last at most N^* concentration windows (Theorem 8.3). Gap-degenerate transits are pancake-type and excluded as in Lemma 6.4; every resolved occupancy exits as a class tube (Lemma 6.4), so transits are counted by their collapse output through the enstrophy-weighted, γ -canceling ledger of Lemma 7.1: at most $C_J + c_r \int \gamma + N_{\text{ep}}$ over the interval, C_J data-only. All factors fold into one exponential. Complete proof: Section 16.2.

Theorem 8.3 (Tangled-horn control). *There are constants M_8 (explicit in the interval data and ν) and ε_8 , N^* , F_{tan} (data-only) such that, under the organization hypothesis and the choice $\varepsilon < \min(\varepsilon_0, \varepsilon_8)$: above M_8 every concentration window at the envelope argmax (the windows of Lemma 9.1; the scale-intensity lock $M_{\text{loc}} \geq (\varepsilon_*/\kappa_3)^{1/2} \nu/s^2$ makes them intensity windows) is quiet — the rescaled local enstrophy vanishes at high intensity, so the window's Duhamel production part obeys $P_1 \leq C_8 \Omega_{\text{sup}}^{1/2} \nu^{-3/4} M^{-1/4} \leq \frac{1}{2}$ — with per-window growth at most $1/(1 - P_1)$ and the viscous persistence floor $\int_{B_R} |\dot{\omega}| \geq c \nu^{3/2}$ (heat-surviving mass at unit rescaled time occupies scale $\sqrt{\nu}$). A sustained-intensity arranged-tangled occupancy therefore recurs, within $N^* = N(\varepsilon_8)$ windows — the covering number of Lemma 9.2's compact, data-rooted class — to a profile carrying the floor in the funnel's own norm, and the rigidity endgame forbids it (Theorem 9.4): persistent arranged-tangled occupancy at sustained intensity is impossible at any point. The total growth factor of any such occupancy is $F_{\text{tan}} \leq e^{N^*}$. Complete proof: Section 17.5.*

Theorem 8.4 (Interval regularity under organization). *Let $0 \leq a < b \leq \infty$ and suppose every episode boundary in $[a, b]$ presents all intense structures class-organized or capped. Then, with the base $B_{a,b} := \max(\|\omega(a)\|_\infty, \sup_{[a,b]} \tilde{r}, M_8)$ — explicit in $(\|\omega(a)\|_\infty, \Omega(a), E_0, \nu, b - a)$ through the interval drift bound, which never references the envelope —*

$$\|\omega(t)\|_\infty \leq B_{a,b} \exp \left[C_{\text{ch}} \left(K(b-a)^{1/6} + K'_2 + K_3(b-a) + K_4(b-a)^{1/2} \right) \right] \quad \text{on } [a, b],$$

where C_{ch} collects the regime rates and the transit and tangled log-factors, $K'_2 = K_2 + C_{\text{CD}}$ plus the counted-transit constants, and $K_4 = c_*(E_0/((2\pi)^3 \nu))^{1/2}$ — every constant explicit. The exact split at $\tilde{r}$: the part below is paid by Theorem 7.3 together with the ω^* -integral, $\int_a^b \omega^* dt \leq K_4(b-a)^{1/2}$ by Cauchy-Schwarz against the energy identity, with no appeal to the linear theory — cores below the clock ratio may be as elliptical as they like; on the part above, every maximizer is intense and load-bearing by construction, provably near-circular in class residence by Lemma 8.1, and rides Lemma 8.2. In particular $\int_a^b \|\omega\|_\infty dt < \infty$ for $b < \infty$, and the Beale-Kato-Majda criterion [BKM84] extends smoothness past b . Complete proof: Section 16.3.

9. THE FUNNEL AND THE RIGIDITY SQUEEZE

Suppose now $T^* < \infty$ is a first singular time. By Theorem 8.4 in its interval form, organization cannot hold on any $[T^* - \delta_0, T^*)$: a sequence of episode boundaries $t_k \uparrow T^*$ carries intense structure outside the class. Let (x^*, T^*) be a singular point. The funnel runs on one quantity — the critical local enstrophy $\mathfrak{E}_{x^*}(s)$ of Section 3.1 — through a dichotomy with teeth on both sides, in the *jet-swept frame*: profiles are compared after subtracting the mean flow and the degree- n Taylor jet of the ball's harmonic exterior field

(the kernel-splitting toolbox of Section 15.3; a finite-dimensional frame, no Lagrangian object). The quotient metric is

$$\tilde{X}([\Phi], [\Psi]) := \inf_Q \|\Phi - \Psi - Q\|_{L^3(B_1)} + \|\nabla \times \Phi - \nabla \times \Psi\|_{L^1(B_1)},$$

the infimum over degree- $\leq n$ polynomial re-frames (the curl component needs no quotient: the rescaled jet curl vanishes in the window limit). Two structural facts shape what follows. The funnel threshold is set in viscous units, $\mathfrak{E}^* := \varepsilon_* \nu^2$. And the *filtering theorem* holds: a flux circuit carrying $\Gamma \geq \Gamma_{\mathfrak{E}} := (\pi \mathfrak{E}^*/2)^{1/2}$ crossing a ball near x^* forces $\mathfrak{E}$ above the threshold there (the circulation floor), so the windows below contain only flux $< \Gamma_{\mathfrak{E}} \ll \Gamma_{\min}$: the chain handles class flux; the funnel sees only what the chain cannot — a theorem, not a design intention. No Lagrangian trajectory enters the localization.

Lemma 9.1 (The concentration dichotomy and the windows). *Define the concentration scale $r_c(t) := \sup\{r \in (0, r_0] : \sup_{0 < s \leq r} \mathfrak{E}_{x^*}(s, t) \leq \mathfrak{E}^*\}$ — well posed ($\mathfrak{E}$ is continuous in (s, t) and vanishes as $s \rightarrow 0$ for smooth fields; the time-slab form runs on backward parabolic cylinders at their own scale). (i) If $\liminf_{t \rightarrow T^*} r_c(t) > 0$ then (x^*, T^*) is regular: the scale-critical strain is $\varepsilon_*^{1/2}$ -small at all small scales (the Riesz shell localization), the vorticity Moser iteration closes in the swept frame for $\varepsilon_* \leq \varepsilon_*^{\text{reg}}$ (the stretching absorbs into dissipation with an $(\varepsilon_*)^2$ -coefficient), and the local Biot–Savart/Serrin criterion concludes. At a singular point concentration is therefore forced: $r_c(t_k) \rightarrow 0$ along boundaries $t_k \uparrow T^*$. (ii) At the realized scales s_k — where $\mathfrak{E}_{x^*}(s_k, t_k) = \mathfrak{E}^*$ exactly, by the intermediate value theorem — the jet-swept window profiles satisfy the local-energy class bounds on every rescaled cylinder Q_R ($\mathfrak{E} \leq \mathfrak{E}^*$ at all sub-window scales by construction; super-cap scales by the race bootstrap of Section 17.5), and carry the exact vorticity floor $\int_{B_1} |\hat{\omega}_k|^2 = \mathfrak{E}^*$ — Galilean-invariant, and jet-invariant in the limit (the rescaled jet curl is $O((s_k/\sigma)^2)$). Complete proof: Section 17.1.*

Lemma 9.2 (The compact class and the recurrence). *The set $\mathcal{K}$ of jet-quotient classes of suitable solutions on Q_2 obeying the window class bounds of Lemma 9.1(ii) with the exact floor is compact in $\tilde{X}$: Aubin–Lions at the velocity and vorticity levels (the local energy inequality in the swept frame; the pressure by the local decomposition; ∂_τ through the equation), with suitability and the floor passing to limits. Its covering number $N(\varepsilon)$ is therefore finite and data-only — the class is equation-constrained and rooted in the data. Among any $N(\varepsilon) + 1$ windows two profiles are ε -close in $\tilde{X}$, and along a subsequence the window profiles recur to a fixed $\Phi_0 \in \mathcal{K}$: the pigeonhole, on an honestly compact class. Complete proof: Section 17.2.*

Lemma 9.3 (The self-similar limit). *Along the recurring subsequence of Lemma 9.2, the scale ratios $\lambda_j = s_{k_{j+1}}/s_{k_j} \in (0, 1)$ either accumulate at some $\lambda \in (0, 1)$ — and the fixed profile is invariant under the parabolic rescaling by λ : the limit is discretely self-similar with factor λ — or tend to 1, giving invariance under a neighborhood of scale factors: exact self-similarity. In either branch the limit is smooth with $|\hat{\omega}| \leq C\varepsilon_*^{1/2}\nu$ at unit normalization, uniformly at every point (off-center $\mathfrak{E}$ -control is inherited: $B_s(y) \subset B_{s+|y|}(x^*)$), ancient, suitable, in the local-energy class — and carries the exact floor $\int_{B_1} |\hat{\omega}|^2 = \mathfrak{E}^*$. Complete proof: Section 17.3.*

Theorem 9.4 (The rigidity endgame). *No such limit exists. In the continuous branch of Lemma 9.3, the rigidity of Nečas–Růžička–Šverák and Tsai [NRŠ96; Tsa98] applies with every hypothesis derived (ancient, suitable, smooth, uniformly small vorticity, local-energy*

class, exactly self-similar), giving $V \equiv 0$; in the discrete branch, the discretely-self-similar rigidity [Tsa14; CW17] applies the same way — the deployment anticipated by v001’s own parenthetical, promoted to the main line. Both contradict the exact floor $\int_{B_1} |\hat{\omega}|^2 = \mathfrak{E}^* > 0$. Hence concentration admits no recurrent limit; with Lemma 9.1(i)’s regular horn, no singular point exists. Complete proof: Section 17.4.

10. PROOF OF THE MAIN THEOREM

Proof of Theorem 1.1. Let $\mathbf{u}_0 \in C^\infty(\mathbb{T}^3)$ be divergence-free and let $\mathbf{u}$ be the unique maximal smooth solution of (1), defined on $[0, T^*)$ with $T^* \in (0, \infty]$ by local existence and uniqueness in the smooth class [Ler34; FK64; Kat84]. The energy identity holds on $[0, T^*)$, so E is non-increasing and $\int_0^{T^*} \Omega dt \leq E_0/2\nu$.

Assume, for contradiction, $T^* < \infty$.

Step 1 (episodes and classification). Partition $[0, T^*)$ into episodes and fix the class constants in the order of Table 1; in particular the funnel threshold $\mathfrak{E}^* = \varepsilon_* \nu^2$ (with $\varepsilon_* \leq \varepsilon_*^{\text{reg}}$, Lemma 9.1) and the recurrence tolerance $\varepsilon < \min(\varepsilon_0, \varepsilon_8)$ (Theorem 8.3) are fixed before this solution is examined. At each episode boundary, Lemma 6.1 classifies every intense point at its active scale, and the organization theorem (Theorem 6.5) leaves exactly three possibilities for each: class-organized, capped, or arranged-tangled.

Step 2 (the organized alternative is regular). Suppose there were $\delta_0 > 0$ such that on $[T^* - \delta_0, T^*)$ every episode boundary presents all intense structures organized or capped. The budget (Theorem 7.3) holds on the interval with its data-only constants; Theorem 8.4 applied on $[T^* - \delta_0, T^*)$ with the smooth data $\mathbf{u}(T^* - \delta_0)$ bounds $\|\omega\|_\infty$ up to T^* ; hence $\int^{T^*} \|\omega\|_\infty dt < \infty$ and Beale–Kato–Majda [BKM84] extends the smooth solution past T^* — contradicting maximality. So no such δ_0 exists.

Step 3 (localization). Therefore a sequence of episode boundaries $t_k \uparrow T^*$ carries intense structure outside the class. Let (x^*, T^*) be a singular point (one exists, else $\|\omega\|_\infty$ is bounded near T^* by compactness and Step 2’s argument applies). At each late episode boundary the classification dichotomy of Lemma 6.1 applies at x^* : coherent-or-capped boundaries chain x^* ’s local intensity through the episode by the envelope principle on the structure’s interior core maximum, against the budget; since $\limsup_{t \rightarrow T^*} |\omega|$ is unbounded at x^* , such boundaries cannot account for all late times — and the concentration dichotomy of Lemma 9.1 has bite at x^* : its regular horn would contradict the singularity, so $r_c(t_k) \rightarrow 0$ and the concentration windows exist.

Step 4 (the funnel). By Lemma 9.1(ii) the concentration windows carry the local-energy class bounds and the exact vorticity floor; by Lemma 9.2 the window profiles recur to a fixed Φ_0 on the compact, data-rooted class; by Lemma 9.3 the limit is exactly (discretely) self-similar, smooth, and uniformly small.

Step 5 (the endgame). By Theorem 9.4 no such limit exists — the rigidity theorems apply with derived hypotheses and contradict the exact floor. This contradiction refutes $T^* < \infty$.

Hence $T^* = \infty$: the solution is smooth on $\mathbb{T}^3 \times [0, \infty)$, unique in the smooth class, with non-increasing energy. Theorem 1.1 is proved. $\square$

Remark 10.1. Every constant invoked above is fixed in Table 1’s dependency order from (E_0, ν) and universal constants alone; every solution-dependent quantity enters only through estimates. The argument runs entirely in physical time; the self-similar variables appear only inside Steps 4–5, in the classical manner of [CKN82; NRŠ96; Tsa98].

11. VERIFICATION APPARATUS

The proof of Theorem 1.1 is the analytic argument of Sections 4 to 10. The apparatus described here is *supporting, not load-bearing*: it motivated the lemmas, adjudicated their exact cores, and cross-checked their mechanisms, but no step of the proof consumes any of it.

11.1. Machine certificates. The exact and algebraic layer of the argument — every closed-form identity, sign, coefficient chain, closure inequality, and evaluated integral the lemmas rest on — is formally verified in Lean 4 over the Mathlib library: **89 kernel-checked theorems, zero unproven placeholders** — the original fifty-four, the sixteen v002 repair certificates (the envelope layer’s arithmetic and the calibrated absorption split), and the eighteen closure-wave certificates (the classification, ledger, race, and funnel algebra; all indexed in Appendix C) — in a project whose Mathlib dependency is pinned and unmodified, reproducible by a single build command. Among them: the financing identity’s complete pointwise algebra including its sign $\text{tr}(SW^2) = +\frac{1}{4}\omega^\top S\omega$; the veto and the parity kernel; the pancake discriminant identity; the modulation Gram pairing $\langle r, W' \rangle = -2$; the calibration constant $\frac{1}{2} - \frac{\ln 2}{3}$ in closed form; the Fujita–Kato beta integral π ; and the closure inequalities of every repair lemma of the final review campaign (the small- ε choice, the logarithmic absorption with its structural coefficient, the circularization threshold, the transit factor, the coherence inheritance, the depletion exponent). The full index — each theorem, what it certifies, and its adjudication log — is Appendix C. A small number of analysis-level steps (a torus integration by parts; two ordinary-differential comparisons; interval-arithmetic eigenvalue enclosures) are named there as declared formalization targets; each is standard, and each is proved in the analytic text.

11.2. Symbolic adjudication. Every algebraic claim passed through a symbolic-computation layer (sympy) before formalization: derivations solved symbolically, identities checked exactly, and every disagreement between the desk mathematics and the machine resolved in writing before the corresponding Lean certificate was attempted. The scripts and their logs are included in the public repository accompanying the paper.

11.3. The instruments. A suite of direct numerical simulations and measurement instruments accompanied the development: coherent-structure identification and tracking on evolved fields, calibrated estimators for core radius and strain (with the analytically exact calibration factor $\frac{1}{2} - \frac{\ln 2}{3}$), alignment and organization statistics (in the tradition of [Ash+87]), and locked-exact tests of the derived scaling laws. Every quoted number sits on a named, calibrated ruler with its tolerance stated; several played a decisive role in *finding* the mathematics (the circularization law of Lemma 8.1, for instance, was calibrated by a measured 1.9:1 elliptical core at clock ratio 19.6 — a measurement that first appeared to threaten the theory and, once the law was derived, confirmed its regime structure). The instruments’ role ends there: the proof stands entirely on the analytic chain. A companion certified-numeric layer — six scripts in the same repository, every infinite tail carrying a proved closed-form majorant and every value rounded outward — archives explicit upper bounds for the named kernel, embedding, and threshold constants of Appendix A; as with every instrument, no step of the proof consumes these values.

11.4. The development record. The complete development record — every derivation, measurement, retraction, and correction, in the order they occurred, including the internal

adversarial review passes over the completed argument, whose findings were each closed in writing at the site of the error — is preserved verbatim by the author and is available upon request.

PART II. THE COMPLETE PROOFS

Part II proves every statement of Part I in full, in the order of the sections that state them; each proof is linked from its statement. The material is ported from the program's authoritative proof versions with their final corrections; nothing here is new relative to Part I — it is the same argument at full length.

12. COMPLETE PROOFS FOR SECTION 4: THE LINEAR THEORY

Throughout, $\Omega(r) = (1 - e^{-r^2})/r^2$, $W(r) = 2e^{-r^2}$, $W' = -4re^{-r^2} < 0$; $\Delta_m = \partial_r^2 + r^{-1}\partial_r - m^2/r^2$ denotes the radial Laplacian in sector m , and the linearized vorticity equation in sector m at spectral parameter c is the Rayleigh system

$$(2) \quad (\Omega(r) - c)\zeta - \frac{W'(r)}{r}\psi = f, \quad \Delta_m\psi = \zeta,$$

with f the (Gaussian-class) datum and the filtered output the pairing of ψ against smooth compactly supported tests.

12.1. Proof of Lemma 4.1. Suppose (c, ψ) solves the homogeneous system (2) with $\text{Im } c \neq 0$ and ψ in the energy class of the column. Since $\Omega - c$ is then nowhere zero, $\zeta = (W'/r)\psi/(\Omega - c)$, and testing $\Delta_m\psi = \zeta$ with $\bar{\psi}$ gives

$$- \int_0^\infty \left(|\psi'|^2 + \frac{m^2}{r^2} |\psi|^2 \right) r dr = \int_0^\infty \frac{W' |\psi|^2}{r(\Omega - c)} r dr.$$

The left side is real; taking the imaginary part of the right side,

$$\text{Im } c \int_0^\infty \frac{W'(r) |\psi(r)|^2}{r |\Omega(r) - c|^2} r dr = 0.$$

Because $W' < 0$ on $(0, \infty)$, the integral is strictly negative unless $\psi \equiv 0$; hence $\text{Im } c = 0$. No sector carries a nonreal eigenvalue. For the three-dimensional statement — spectral stability of the columnar vortex under perturbations with no symmetry restriction — see [GS20], whose hypotheses the Oseen column satisfies. $\square$

12.2. Proof of Lemma 4.2. Fix $m \geq 2$ and Gaussian-class f . We bound the filtered solution of (2) uniformly for c in the closed band $[0, \Omega(0)]$ and in the regularization $c \mapsto c + i0$.

(a) *Monotone reduction and the critical layer.* Ω is strictly decreasing ($\Omega' < 0$ on $(0, \infty)$, $\Omega(0^+) = 1$, $\Omega(\infty) = 0$), so $v = \Omega(r)$ is a diffeomorphism of $(0, \infty)$ onto $(0, 1)$. In the variable v the singular factor of (2) is $(v - c)^{-1}$, and the limiting value as $\text{Im } c \downarrow 0$ is the Plemelj boundary operator $\text{P.V.}(v - c)^{-1} + i\pi\delta(v - c)$. On the Jacobian-weighted space $X_w = L^2(w dv)$, $w(v) = |\Omega'(\Omega^{-1}(v))|^{-1}$, the principal-value part is the Hilbert transform up to the bi-Lipschitz change of variables and is bounded with the Riesz constant; the trace part is bounded by the trace inequality on X_w . Neither bound depends on m .

(b) *The annulus Green's function.* On any annulus $0 < r_1 \leq r \leq r_2 < \infty$ the operator Δ_m has homogeneous solutions $r^{\pm m}$ and Green's function

$$G_m(r, s) = -\frac{1}{2m} \left(\frac{r_{<}}{r_{>}} \right)^m, \quad r_{<} = \min(r, s), \quad r_{>} = \max(r, s),$$

by the Wronskian normalization $W[r^m, r^{-m}] = -2m/r$. The prefactor $1/2m$ and the decay $(r_{<}/r_{>})^m$ are the two m -improving factors consumed by Lemma 4.4.

(c) *Core depletion, with the derived exponent.* Near the axis $\Omega(r) = \Omega(0) - \frac{1}{2}|\Omega''(0)|r^2 + O(r^4)$ with $\Omega''(0) = -1$ exactly, so for c close to the core frequency the critical radius

satisfies $r_c = (2(\Omega(0) - c)/|\Omega''(0)|)^{1/2}(1 + o(1))$, and $|\Omega'(r_c)| = |\Omega''(0)| r_c(1 + o(1))$. Both the response and the sector component of smooth data vanish centrifugally at the axis — $\psi, f_m \sim r^m$ — so the resonant trace obeys

$$|\text{trace}(c)| \leq C \frac{r_c^{2m}}{|\Omega'(r_c)|} \leq C_d (\Omega(0) - c)^{m-\frac{1}{2}},$$

a positive power for every $m \geq 1$: the resolvent stays bounded up to the core edge. (The exponent $m - \frac{1}{2}$, not m , is the honest output of the quadratic profile; the conclusion needs only positivity.)

(d) *The outer edge.* As $c \downarrow 0$ the critical layer moves to $r_c \rightarrow \infty$, where every source is Gaussian-small: the datum and the coupling W'/r carry the factor $e^{-r_c^2}$, i.e. $e^{-\Theta(1/c)}$ in the band variable, while the layer weight w grows only polynomially, $O(c^{-3/2})$. The product vanishes at the edge, and the Green's decay $(r_{<}/r_{>})^m$ sums the exterior contribution.

Assembly. Split the band into the core neighborhood, the outer edge, and the compact interior. On the interior the critical radius ranges over a compact set on which $|\Omega'| \geq c_0 > 0$, and (a)+(b) bound the limiting resolvent; (c) and (d) bound the two edges; uniformity in the regularization follows since every estimate was taken at the boundary value. $\square$

12.3. Proof of Lemma 4.3. *The kernel, exactly.* With $V(r)$ the column's velocity profile ($V = r\Omega(r)$), direct differentiation gives $\Delta_1 V = V'' + r^{-1}V' - r^{-2}V = W'$ and $\Omega W' - (W'/r)V = W'(\Omega - V/r) \equiv 0$: the pair $(\zeta_T, \psi_T) = (W', V)$ solves the homogeneous system (2) at $c = 0$ — the infinitesimal translation of the column.

The adjoint kernel and the pairing. The impulse $\int_{\mathbb{R}^2} x \times \omega \, dx$ is conserved by the linearized flow, and its density in sector $m = 1$ is the functional $\varphi(r) = r$. The pairing is exact:

$$\langle r, W' \rangle = \int_0^\infty W'(r) r^2 \, dr = \left[W r^2 \right]_0^\infty - 2 \int_0^\infty W r \, dr = -2 \int_0^\infty 2e^{-r^2} r \, dr = -2,$$

(machine-certified as `gram_integral`); the zero-frequency pole is therefore simple and well conditioned: the spectral projection onto the translation pair is bounded with constant $1/2$.

No embedded neutral modes. A neutral mode at $c \in (0, \Omega(0))$ would require $W'(r_c) = 0$ at its critical radius; $W' < 0$ excludes it.

Zero-impulse bounds. On data with $\langle r, f \rangle = 0$ the resolvent acts on the complement of the generalized kernel; the mechanisms (a)–(d) of Section 12.2 apply verbatim in $m = 1$ (the depletion exponent $m - \frac{1}{2} = \frac{1}{2}$ remains positive), giving the full-band bound. The impulse component itself is the free drift of the center, which the nonlinear theory of Section 5 removes as a modulation coordinate rather than inverting. $\square$

12.4. Proof of Lemma 4.4. Each m -dependence in Section 12.2 improves with m : the Green's operator built on G_m has norm at most $\frac{1}{2m} \sup_{r>0} \int_0^\infty (r_{<}/r_{>})^m \frac{ds}{s} \leq \frac{C}{m}$ (the substitution $x = r_{<}/r_{>}$ gives $\int_0^1 x^m dx/x$ -type integrals; in the normalized form $\int_0^1 x^m dx = 1/(m+1)$, machine-certified as `su_green_integral`); the Plemelj constant of (a) never sees m ; the depletion exponent $m - \frac{1}{2}$ of (c) strengthens with m ; and the centrifugal term gives $\|\Delta_m^{-1}\| \leq C/m^2$ on the weighted class. Hence $C(m) \leq C_{\text{su}}/m$, and for smooth data, whose sector norms decay rapidly, $\sum_m C(m) \|f_m\| < \infty$. $\square$

12.5. Proof of Theorem 4.5. *The matched approximation.* Fix $m \geq 2$ and c in the band, and let ψ_0 be the inviscid boundary-value solution of Section 12.2. Across the critical radius the viscous equation replaces the Plemelj jump by the Airy-regularized transverse model

$i\lambda v\zeta - \nu\zeta'' = f$ (a Scorer-type layer of width $\nu^{1/3}$), whose solutions converge to the Plemelj boundary value with error $O(\nu^{1/3})$ on C^1 data, uniformly on the band (uniform asymptotics for the layer functions as in [Olv74]). Choose the matching radius $\ell = \nu^{1/6}$ and define the additive composite

$$\psi_{\text{app}} = \psi_0 + \chi_\ell (\psi^{\text{lay}} - \psi^{\text{frozen}}),$$

where ψ^{lay} is the layer solution, ψ^{frozen} its frozen-coefficient inviscid limit, and χ_ℓ a cutoff at scale ℓ ; the additive form corrects the layer without disturbing the regular part. The residual of ψ_{app} is supported in the layer and bounded by $C\nu^{1/3}$ in the filtered norm: the two binding terms are the layer's curvature correction ($\ell^2 = \nu^{1/3}$) and the Plemelj mismatch ($O(\nu^{1/3})$ above).

The parametrix iteration. The residual class — layer-profile functions plus Gaussian-smooth remainders — is invariant under the matched solve; one pass costs $O(1)$ in filtered output and returns a residual smaller by the factor $C\nu^{1/3}$: the $\nu^{-1/3}$ inversion cost of the layer is exactly canceled by the layer datum's integrated smallness $O(\nu^{1/3})$ (width times height). For $\nu \leq \nu_0$ the iteration converges geometrically, giving $\sup_\nu \sup_c \|\psi_{\nu,c}\| \leq C(m)$ and $\|\psi_\nu - \psi_0\| \leq C'\nu^{1/3}$. $\square$

12.6. The three-dimensional transfers. For a columnar vortex with no axial base flow, the critical-layer condition in every three-dimensional sector (m, q) is still $m\Omega(r_c) = s$: the axial wavenumber q enters neither the layer's location nor its transverse model, so Theorem 4.5's layer analysis transfers verbatim. What q changes is the outer operator: Rayleigh's is replaced by the Howard–Gupta form, whose annulus Green's function retains the $(r_</r_>)$ -type decay and whose core behavior retains the centrifugal vanishing, so mechanisms (b)–(d) of Section 12.2 have direct analogues; the centrifugal stability of the column under general three-dimensional perturbations is [GS20]. The sectors $(m, q) = (0, q \neq 0)$ carry inertial-wave turning points in place of a critical layer; they have no stretching coupling and are controlled by the same weighted energy estimate as Section 12.1. The downstream consumers (Section 5) use exactly these transferred bounds. $\square$

12.7. Proof of Lemma 4.6. *The frame and the split.* The class column is Burgers-strained ((C1): $\delta^2 = 4\nu/\alpha$), and in the strained core frame with the Gaussian weight ρ the linearized vorticity operator splits as $L = A + R + B$: A the Fokker–Planck part (viscosity plus strain advection), self-adjoint in $L^2(\rho)$ with compact resolvent and discrete spectrum, gap a multiple of α on the non-kernel sectors (the same classical device as Section 13.11's mode architecture); R the differential rotation $\Omega(r)\partial_\theta$, *exactly skew* in $L^2(\rho)$: the rotation field is divergence-free and tangent to the level sets of the radial weight, so

$$\int \rho v (u_{\text{rot}} \cdot \nabla v) = \frac{1}{2} \int \rho u_{\text{rot}} \cdot \nabla (v^2) = -\frac{1}{2} \int v^2 \nabla \cdot (\rho u_{\text{rot}}) = 0$$

(an exact identity: $u_{\text{rot}} \perp \nabla \rho$ and $\text{div } u_{\text{rot}} = 0$; machine-certified, `r6_skew_divergence`); and B the coupling of the perturbation velocity against the base gradients, bounded on the class family with norm continuous in the parameters.

(i) *Boundedness.* A is sectorial, R skew, B bounded, so L generates an analytic semigroup [EN00]; its bound on the constrained subspace is continuous over the compact family, hence uniformly finite.

(ii) *The $m \geq 2$ gap.* No unstable or neutral modes exist on the sector: a nonreal eigenvalue is excluded by the Rayleigh sign identity (Section 12.1, whose sign argument the viscous part only strengthens); a purely imaginary eigenvalue is excluded for the Gaussian profile by

the imaginary-axis resolvent bound of [LWZ20], and the (C1)-margin perturbation moves the discrete spectrum continuously and cannot reach the axis within the class margins. The spectrum is discrete (compact resolvent) and strictly left of the axis; the spectral mapping theorem for analytic semigroups [EN00] gives decay at the spectral abscissa; the abscissa is continuous on the family (analytic perturbation of discrete spectra), hence its supremum over the compact family is negative: the class floor $c_2 > 0$.

(iii) *The three-dimensional sectors.* The column has no axial base flow, so the advection frequencies are $m\Omega(r)$ independently of q , the stretching coupling vanishes in the $(0, q)$ sectors and is class-margin-bounded elsewhere, and the weighted energy estimate closes boundedness exactly as in two dimensions; no gap is claimed and none is consumed.

The demotion. The consumer audit: Section 13.11 consumes (i) (the adiabatic bound M); Section 14.3 consumes the $m = 1$ band and, through Lemma 8.1, (ii)'s floor ($C_e = C_f/c_2$ and $R_7 = 2MC_e$ need only $c_2 > 0$ and $M < \infty$); nothing consumes the $\nu^{1/3}$ form or the transfer sharpness. $\square$

13. COMPLETE PROOFS FOR SECTION 5: THE EPISODE MACHINERY

13.1. The filtered frame. Fix a class tube at an episode boundary. In core units the solution near the tube is written

$$\mathbf{u} = U_{\sigma(t)} + v,$$

where U_{σ} is the Oseen column at modulation parameters $\sigma = (X_1, X_2, \delta^2, \Gamma)$ — transverse center, squared core radius, circulation — and v is the perturbation, measured in the Gaussian-weighted norms $\|\cdot\|_F = \|\rho^{1/2} \cdot\|_{L^2}$ and its one-derivative companion $\|\cdot\|_{F_1}$, ρ the core Gaussian. The secular directions are removed by the exact adjoint functionals of Section 12.3: v is constrained orthogonal to the translation pair (twice) and to the dilation and circulation densities. All constants in this section depend only on the class constants of Table 1; the episode has length c/γ , and the hypotheses (C1)–(C2) hold at its boundary.

13.2. The far-field kernel bound. Let Q be a vorticity region at distance $d \geq R_m\delta$ from the tube. Writing the strain as the second gradient of the Biot–Savart potential and integrating by parts once to the velocity,

$$|S[\omega \mathbf{1}_Q](x)| \leq C_K \int_Q \frac{|\mathbf{u}(y)|}{|x-y|^4} dy \leq C_K \left(\int_Q |x-y|^{-8} dy \right)^{1/2} \|\mathbf{u}\|_{L^2(Q)},$$

by Cauchy–Schwarz. For Q exterior to the ball of radius d the radial integral is $\int_d^\infty s^{-8} s^2 ds \cdot |\mathbb{S}^2| = \frac{4\pi}{5} d^{-5}$ (the tail evaluation $\int_d^\infty s^{-6} ds = d^{-5}/5$ is machine-certified, `w8.tail.integral`; its one-derivative companion $\int_d^\infty s^{-4} ds = d^{-3}/3$ is `w1.tail.integral`), giving the far-field display

$$|S[\omega \mathbf{1}_Q](x)| \leq \left(\frac{4\pi}{5} \right)^{1/2} C_K d^{-5/2} \|\mathbf{u}\|_{L^2(Q)},$$

and the companion gradient bound with constant $(\frac{7\pi}{6})^{1/2}$ by the same computation one derivative higher. $\square$

13.3. The parity law. Let the tube's centerline have unit tangent ξ with geometry gradients $\nabla \xi, \nabla^2 \xi$. The near-field contribution of the tube's own vorticity to its core-averaged axial strain is a multilinear functional of (ω, ξ) whose kernel is *odd* under reflection through the core plane at leading order (machine-certified, `parity.kernel.odd`): the term linear in the

displacement integrates to zero against the even core profile — uniform bending does not stretch. The first surviving terms carry two geometry gradients or one gradient of $|\omega|$:

$$|\alpha_{\text{near}}| \leq C r^2 \left[(\|\nabla^2 \xi\| + \|\nabla \xi\|^2) \sup |\omega| + \|\nabla \xi\| \sup |\nabla \omega| \right],$$

with r the core radius and the constant $\frac{3}{2}$ in the moment form of the display (the exact multilinearity replaces any Lipschitz surrogate: the identity is polynomial in the displacement field, and the parity kill is exact). Under (C1) ($\kappa\delta \leq 1/M$) the bracket is a class-bounded multiple of the slow-clock scale. $\square$

13.4. The episode gain. The two resonant amplitudes $a = (a_+, a_-)$ of the swept normal form satisfy $\dot{a} = (L_0 + \varepsilon K(t))a + F(t)$ with L_0 skew ($\text{Re}\langle a, L_0 a \rangle = 0$ exactly; the $\mathbb{C}^2$ -skew content is machine-certified, `skew_core`, `hermitian_quadratic_real`) and εK the strain coupling. The log-norm method gives $\frac{d}{dt}\|a\| \leq \varepsilon \varsigma \|a\| + \|F\|$ (assembly `w3_lognorm_assembly`), hence over an episode of length T

$$\|a(T)\| \leq e^{\varepsilon \varsigma T} (\|a(0)\| + T \sup \|F\|),$$

order one per episode; and the frequency sweep caps the resonant residence, the swept Grönwall closing with the factor $\exp(2\varsigma^2/w)$, w the sweep rate. $\square$

13.5. Modulation well-posedness. The secular constraints $\langle v, \varphi_j \rangle = 0$ define the modulation map $\sigma \mapsto v = \mathbf{u} - U_\sigma$ implicitly. Its linearization at $v = 0$ is the Gram matrix of the kernel densities against the adjoint functionals,

$$\text{Gram} = \text{diag}(-2, -2, 1, 1),$$

whose translation entries are the exact pairing $\langle v, W' \rangle = -2$ of Section 12.3 (machine-certified, `gram_integral`) and whose dilation and circulation entries are the normalized moments, exactly 1. The matrix is nondegenerate; the Banach implicit function theorem gives, for $\|v\|_F$ below a class threshold, unique modulation parameters depending smoothly on the solution, with the modulation equations

$$\dot{\sigma} = V(\sigma) + \mathcal{N}(\sigma, v), \quad |\mathcal{N}| \leq C \|v\|_F^2 :$$

the drift of center, radius, and circulation is slaved to the perturbation at quadratic order, and the secular projections hold exactly along the flow. $\square$

13.6. The adiabatic estimate. Let $S(t, s)$ be the evolution of the swept linear operator $L(t)$ on the constrained subspace. Freezing coefficients on a window of length τ and expanding by Duhamel, the frozen evolution is bounded by M (the spectral bound of the frozen operator on the constrained subspace, from Sections 12.2 and 12.3), and the commutator cost of the sweep over the window is $M\Lambda\tau^2/2$ with Λ the sweep rate. At the threshold

$$\Lambda \leq \frac{\mu^2}{4M(1 + \ln M)^2}$$

the window length $\tau = 2(1 + \ln M)/\mu$ makes the Grönwall exponent of the frozen error exactly $\frac{1}{2}$ (machine-certified, `w5_boundary_exponent`) while the damping $e^{-\mu\tau/2}$ gains the factor $e^{-(1+\ln M)}$: chaining windows,

$$\|S(t, s)\| \leq M e^{-\frac{\mu}{2}(t-s)},$$

with the window slack exactly $\frac{1}{2}$ per window (`w5_window_slack`). In class units the threshold reads $\gamma/\omega_0 \leq c_1^2/(4MC_L(1 + \ln M)^2)$ — comfortably satisfied in the load-bearing regime $\omega_0 \geq R_7\gamma$. $\square$

13.7. The bilinear estimate and the bootstrap root. For the quadratic term $N(v, w)$ in the filtered scale, the duality product rule is exact: testing against $\rho\phi$ and moving one derivative, $|\langle N(v, w), \phi \rangle_F| \leq C\|v\|_F\|w\|_{F_1}\|\phi\|_{F_1}$, and interpolation against admissible tests gives $\|N(v, w)\|_F \leq C_{\text{eq}}\|v\|_F\|w\|_{F_1}$. Combined with the smoothing of Section 13.8 this closes the quadratic loop: the episode’s a-priori bound solves $y \leq \eta + C_1 y^2$, whose lower root obeys $y \leq \frac{1}{2C_1}(1 - \sqrt{1 - 4C_1\eta}) \leq 2\eta$ by the root bound $1 - \sqrt{1 - s} \leq s$ (machine-certified, `w6_closure`): smallness propagates across the episode below the explicit threshold. $\square$

13.8. Smoothing, uniformly over the episode. Write $L[\sigma] = \nu\Delta + B[\sigma]$ with B the first-order Oseen coupling, and let $\mathcal{K}$ be the episode’s compact parameter set ((C1)–(C2) bounds). The Gaussian-weighted heat kernel obeys $\|\nabla e^{\nu\tau\Delta} f\|_{L_\rho^2} \leq C_h(\nu\tau)^{-1/2}\|f\|_{L_\rho^2}$ (the commutator $[\nabla, \rho]/\rho$ is a bounded multiplier), and Duhamel absorbs B : the first-order cost is paid by the kernel’s $(\nu(\tau - s))^{-1/2}$ with $\int_0^\tau (\nu(\tau - s))^{-1/2} ds = 2(\tau/\nu)^{1/2}$, the iteration converging geometrically for

$$\tau \leq \tau_0 = \min\left(\frac{1}{\mu}, \frac{\nu}{4b^2C_h^2}\right), \quad b = \sup_{\sigma \in \mathcal{K}} \|B[\sigma]\|_{C_\rho^1} < \infty,$$

yielding $\|e^{\tau L[\sigma]} v\|_{F_1} \leq C_s \tau^{-1/2} \|v\|_F$ with $C_s = 2C_h C_{\text{eq}} \nu^{-1/2}$, uniformly over $\sigma \in \mathcal{K}$ by compactness. $\square$

13.9. Local existence in the filtered scale. The integral equation $v(t) = S(t, 0)v_0 + \int_0^t S(t, s)[N(v, v) + F_{\text{ext}}] ds$ is a contraction on a ball of $X_T = C([0, T]; F) \cap \{t^{1/2}v \in C((0, T]; F_1)\}$: the linear part is bounded by Sections 13.6 and 13.8, the bilinear part by Section 13.7, and the only integral in the bookkeeping is exact,

$$\int_0^t (t - s)^{-1/2} s^{-1/2} ds = \pi$$

(machine-certified with its three supporting theorems, `w16_beta_integral`). Banach’s fixed point gives existence and uniqueness for $T \leq T_0$ explicit; continuation holds while $\|v(t)\|_F$ is finite; and while the smooth solution of (1) exists, uniqueness in X_T identifies the filtered solution with its restriction (the Fujita–Kato scheme [FK64; Kat84] instantiated in the weighted frame). $\square$

13.10. Spectral slaving of the exterior. The tube’s exterior field is slaved to the core: the derivative of the exterior multipole response to a core mode decays as $(\frac{4\pi}{7})^{1/2} r^{-7/2}$ (the certified tail evaluations of this kernel family are $\int_r^\infty s^{-4} ds = r^{-3}/3$ and $\int_r^\infty s^{-6} ds = r^{-5}/5$ — `w1_tail_integral`, `w8_tail_integral`), and the m -th multipole carries the sector factor $(\delta/R)^m$ at exterior radius R : exterior structure cannot feed back onto the core except through the ambient strain already counted in the budget. This is the far-field half of hypothesis (C2)’s re-supply. $\square$

13.11. Proof of Theorem 5.1. Let the class tube satisfy (C1)–(C2) at the episode boundary.

Step 1 (the frame exists). By Section 13.9 the filtered local problem has a unique solution on the episode, identified with the restriction of the smooth flow; by Section 13.5 the modulation parameters are well defined and the secular projections hold along it.

Step 2 (the forcing is slow-clock). The external forcing of the local problem is the ambient strain: the far field by Section 13.2 summed as in Section 7, the tube's own bending by the parity law Section 13.3 under (C1), and the exterior feedback by Section 13.10. Total: $\|F_{\text{ext}}\|_F \leq C\gamma$.

Step 3 (the linear evolution damps). On the constrained subspace the swept operator satisfies the adiabatic bound $\|S(t, s)\| \leq Me^{-\mu(t-s)/2}$ (Section 13.6), the resonant pair contributing order one per episode under the sweep (Section 13.4).

Step 4 (nonlinear closure). With Steps 2–3, Duhamel and the bilinear estimate close the quadratic loop at the root bound (Sections 13.7 and 13.8): for episode data with $\|v(0)\|_F \leq c^*$ — supplied by clause (C1'); fresh exits reach the seed after $k_{\text{grace}} = \lceil \log(2C_{\text{ent}}/c^*)/(2c_1c) \rceil$ boundaries, their grace residence charged to the transit ledger — the perturbation obeys $\|v(t)\|_F \leq C\gamma/\mu$ throughout the episode; the maximal-time bootstrap extends this to the whole episode since the bound re-verifies its own hypothesis below c^* .

Step 5 (the cap). The core-averaged stretching rate splits as $\alpha = \alpha[U_\sigma] + \alpha[v] + \alpha_{\text{amb}}$. The column's own axial strain vanishes identically (the veto, `veto_algebra`); the perturbation contributes $|\alpha[v]| \leq C\|v\|_{F_1} \leq C'\gamma$ by Step 4 and smoothing; and the ambient term is $\leq C\gamma$ by Step 2. Hence $\alpha(t) \leq C'_1\gamma(t)$ throughout the episode, with C'_1 explicit in the class constants. The hypotheses for the next episode are re-supplied as stated (organization by Theorem 6.5; spectral inputs by Lemmas 4.2 to 4.4 and 4.6 and Theorem 4.5). $\square$

14. COMPLETE PROOFS FOR SECTION 6: CLASSIFICATION

14.1. Proof of Lemma 6.1. (i) *The active scale exists and is measurable.* For each intense point x the set of coherent dyadic scales is determined by countably many measurable conditions ($S_{\text{ext},r}(x)$ is jointly continuous in (x, r) for a smooth field — the kernel bounds with the smooth cutoff — so the coherence and floor conditions at each dyadic scale are measurable), and $r(x) = \max\{r_j : B_{r_j}(x) \text{ coherent}\}$ is well defined and measurable wherever some scale qualifies.

(ii) *Downward inheritance, relative to the parent organizer.* On the dyadic child $B_{r'}$, $r' = r/2$, the two exterior fields differ by the strain of the annulus sources: $S_{\text{ext},r'} - S_{\text{ext},r} = S[\omega(\zeta_{r'} - \zeta_r)]$, supported in $B_{3r} \setminus B_r$ at distance $\geq r/2$ from $B_{r'}$, so

$$\begin{aligned} \sup_{B_{r'}} |S_{\text{ext},r'} - S_{\text{ext},r}| &\leq C'_K (r/2)^{-3} \|\omega\|_{L^1(B_{3r})} \\ &\leq C_{\text{ann}} r^{-2} \mathfrak{E}_{x,\text{sup}}(3r)^{1/2}, \quad C_{\text{ann}} := 24 \kappa_3^{1/2} C'_K, \end{aligned}$$

with $\kappa_3 = 4\pi/3$ and C'_K the named classical kernel-derivative constant. Under the organizer floor and the clause $\mu_0 \geq 2C_{\text{ann}}/\Delta_0$ this is at most $(\Delta_0/2)\|\bar{S}_{\text{ext},r}\|$; the child's oscillation about the parent organizer is then at most $2\Delta_0\|\bar{S}_{\text{ext},r}\|$ while the child mean obeys $\|\bar{S}_{\text{ext},r'}\| \geq (1 - \frac{3}{2}\Delta_0)\|\bar{S}_{\text{ext},r}\|$: the relative oscillation is at most $2\Delta_0/(1 - \frac{3}{2}\Delta_0) \leq 3\Delta_0$ — exactly where $\Delta_0 \leq \frac{1}{6}$ enters $(2/(1 - \frac{1}{4}) = \frac{8}{3} \leq 3$; the arithmetic is machine-certified in the closure wave, `s1_inheritance`; the superseded full-strain algebra `dc_coherence_inherit` remains in the tree, true, role demoted). The child's own floor is *not* claimed — the floor strengthens at

smaller scales and no estimate below consumes it. Every growth estimate runs with the largest coherent organizer — the strain actually driving the point.

(iii) *The covering.* The active-scale balls $\{B_{r(x)}(x)\}$ cover the intense set; the Besicovitch covering theorem [Mat95] extracts a countable subfamily of overlap at most the dimensional constant b_3 . All counts, enstrophy weights, and budget sums of Section 7 run over this subfamily, each ball classified once.

(iv) *No active scale.* If no dyadic scale qualifies for x at any window of a sequence of episode boundaries, then along those windows the strain about x supplies no organizer at any scale $r_j = 2^{-j}r_0$: each scale is either Δ_0 -incoherent or organizer-starved; in the starved mode the exterior paces the pairing at the margin scale $(1 + \Delta_0)\mu(r_j)$ and Definition 6.2's test caps the point, so a point that sustains production presents the arranged-tangled horn's defining state. The funnel's input at such a point (the concentration windows and the profile recurrence) is *derived* from this state and the singular-point dichotomy in Section 9; nothing is consumed by translation. $\square$

14.2. Proof of Lemma 6.3. Let S be the organizing strain of a coherent patch, with unit eigenframe (e_1, e_2, e_3) , eigenvalues $\lambda_1 \geq \lambda_2 \geq \lambda_3$, and gap $g = \lambda_1 - \lambda_2$ in appropriate units; let Δ_0 bound the patch incoherence and ρ the frame drift along trajectories. The direction $\xi = \omega/|\omega|$ obeys

$$D_t \xi = S\xi - (\xi^\top S \xi) \xi + R(t)\xi, \quad \|R\| \leq C(\Delta_0 + \rho),$$

the projective strain flow plus the incoherence/drift perturbation. Write $\tan \theta$ for the tangent of the angle between ξ and e_1 . In the eigenframe the projective flow gives, for the component along any e_k ($k \geq 2$), $D_t(\xi_k/\xi_1) = -(\lambda_1 - \lambda_k)(\xi_k/\xi_1) + O(\Delta_0 + \rho)$, hence

$$\frac{d}{dt} \tan \theta \leq -g \tan \theta + C(\Delta_0 + \rho), \quad \tan \theta(t) \leq e^{-gt} \tan \theta(0) + \frac{C(\Delta_0 + \rho)}{g} :$$

alignment contracts on the organizer's own clock to the equilibrium level $\theta_{\text{eq}} \leq C(\Delta_0 + \rho)/g$.

For the pinning bound, suppose the point sustains pairing above the cap: $\xi^\top S \xi \geq \lambda_2 + m$ on the trailing window. Expanding in the eigenframe, $\xi^\top S \xi = \lambda_1 - [(\lambda_1 - \lambda_2)\xi_2^2 + (\lambda_1 - \lambda_3)\xi_3^2]$, so

$$(\lambda_1 - \lambda_2)\xi_2^2 + (\lambda_1 - \lambda_3)\xi_3^2 \leq \lambda_1 - \lambda_2 - m + \Delta_0 = g - m + \Delta_0,$$

(the Δ_0 absorbing the patch incoherence of the eigenvalues), and since $\sin^2 \theta = \xi_2^2 + \xi_3^2$,

$$\sin^2 \theta \leq (g - m + \Delta_0) \left(\frac{1}{g} + \frac{1}{\lambda_1 - \lambda_3} \right) :$$

sustained pairing pins the direction into the alignment cone. $\square$

14.3. Proof of Lemma 6.4. Let the coherent family be aligned as in Section 14.2, with core intensity $c\omega_0$ and residual misalignment at most θ^* . Decompose the strain at the family's axis into its three sources.

The far field contributes at most γ_r , the budget entry of Section 7, by definition.

The family's own bending contributes through the parity law: by Section 13.3 in moment form, the self-induced axial strain of a curved, core-concentrated family is bounded by $\frac{3}{2}r^2[(m_2 + m_1^2)\Omega_0 + m_1g_1]$, where m_1, m_2 are the first and second parity moments of the core profile and Ω_0, g_1 the core rotation and its gradient scale; for the near-Gaussian class profile the odd moment m_1 vanishes at leading order and m_2 is the certified second moment — the bracket is a class-bounded multiple of the slow-clock scale under (C1).

The *residual misalignment* contributes quadratically: by the no- $\cos^2 \theta$ expansion of the pairing against a field whose axial diagonal vanishes (machine-certified, `cp_axial_residue`, with `veto_algebra` killing the aligned part), the surviving *instantaneous* term is $\leq C\theta^{*2}c\omega_0$. It does not persist: the tilt lives in the $m = 1$ sector, whose zero mode is removed by the modulation and whose remaining spectrum has no neutral modes (Lemmas 4.1 and 4.3: $W' < 0$; the constrained tilt oscillates at frequencies $\geq c_f\omega_0$), while the drive varies at the ambient rate; the 2β -harmonic residual is a pure oscillation, $\theta^\top S_\perp \theta = |\theta|^2(\tau/2)\cos(2\varphi - 2\beta)$ (machine-certified, `r4c_zero_mean`), so one integration by parts on the trailing window $\tau_w = c_w/\gamma$ gives the averaged bound $\langle \alpha_{\text{res}} \rangle_{\tau_w} \leq C_{\text{av}}\theta^{*2}\gamma$, C_{av} explicit in (c_w, c_f) and the episode drift constants.

Summing, in window-averaged form,

$$\langle \lambda_{\text{axis}} \rangle_{\tau_w} \leq \gamma_r + \frac{3}{2}r^2[(m_2 + m_1^2)\Omega_0 + m_1g_1] + C_{\text{av}}\theta^{*2}\gamma.$$

With the θ^* -clause $\theta^{*2} \leq m/(2C_{\text{av}})$ and $c\omega_0 \gg \gamma_r$ (the clock-ratio regime of Table 1), the right side falls below the cap $\lambda_2 + m$ at *every* intensity: the point can no longer sustain window-averaged pairing — it exits to the capped horn. The exiting structure is coherent (the alignment cone), core-concentrated (the Gaussian frame), and carries circulation bounded below by the *derived* floor: the aligned component of ω through the half-radius core disc is $\geq \cos \theta^* \cdot (\omega_0/2)$ on the class profile's guaranteed fraction, so $\Gamma \geq c_{\text{cc}}\omega_0\delta^2 \geq (4c_{\text{cc}}R_7/C_1')\nu = \Gamma_{\text{min}}$ — the γ 's cancel exactly (machine-certified, `r3a_gamma_min`); no conservation claim enters, and the floor is re-supplied at every boundary: clauses (C1)–(C2) hold, and the exit is a class tube.

The pancake fork. The collapse used $g > 0$. The degenerate boundary $g \rightarrow 0$ with λ_1 of order one is the pancake: it is the equality case of the discriminant inequality $54(\lambda_1\lambda_2\lambda_3)^2 \leq (\sum \lambda_i^2)^3$ (machine-certified, `pancake_extremal_sq`), i.e. two leading eigenvalues coalescing with the third balancing. Under the 3:1 deformation law a gap-degenerate family contracts into the top eigenplane — a sheet family. A sheet family persists to a singular time only at unbounded intensity, so every point of the limiting sheet carries unbounded vorticity in each of its parabolic neighborhoods — every such point is singular, and the sheet places a two-dimensional set inside the singular set, contradicting the partial-regularity theorem [CKN82; Lin98], whose singular set has vanishing one-dimensional parabolic measure and a fortiori contains no surface. The pancake branch therefore rejoins the excluded alternatives, and the collapse covers every organizing configuration. $\square$

14.4. Proof of Theorem 6.5. Fix an episode boundary and an intense point x , classified at its active scale by Section 14.1.

Case 1: organizing. The active ball is coherent with some gap g . If g is nondegenerate, Section 14.2 aligns the point within the fast time $T_c = g^{-1}\log(1/\theta^*)$ and Section 14.3 then collapses the aligned eigenvalue at intensity: the point exits to the capped horn carrying tube structure (C1)–(C2). If the gap degenerates, the pancake fork of Section 14.3 excludes persistence. Either way, an organizing point that sustains production becomes a class tube (or is capped) within the fast clock.

Case 2: capped. By Definition 6.2 the trailing-window average of the pairing is at most $\lambda_2 + m$; at the cap's margin the production is $\leq (\lambda_2 + m)$ -paced, and λ_2 at tube-dominated points is the vetoed axial eigenvalue (zero at leading order, `ft_middle_eigenvalue`) while at background-dominated points it is the budgeted quantity of Lemma 7.2: capped points are productionless up to the slow clock and the counted margin.

Case 3: arranged-tangled. By Section 14.1(iv) the point satisfies the multi-window maintenance hypothesis — exactly the funnel’s input, excluded in Section 9 (Theorem 9.4).

Measurability and bookkeeping. Each horn is defined by measurable conditions (time-averages of continuous quantities; sups over dyadic scales), so the horns partition the intense set measurably; the Besicovitch subfamily of Section 14.1(iii) carries the counts; and every exit, culling, and merger event is logged by the episode ledger of Section 7, whose costs are priced there. Consequently at every episode boundary each intense structure sustaining production is class-organized or capped-productionless, and the hypotheses of Theorem 5.1 are re-supplied. $\square$

15. COMPLETE PROOFS FOR SECTION 7: THE STRAIN BUDGET

15.1. Proof of Lemma 7.1. (i) *The vetoed pair kernel.* Let two distinct class tubes have separation $d \geq R_m \max(\delta_i, \delta_j)$ (clause (C3): closer pairs are one structure), axis angle θ_{ij} , and let κ_j bound the companion’s curvature. Near the closest-approach segment, model the companion by its local filament and write its transverse displacement from the straight parallel line as $h(z)$, $|h(z)| \leq \frac{1}{2}\kappa_j z^2$ on $|z| \lesssim d$. To first order in h the companion’s velocity field is the *shifted two-dimensional field*

$$u(x_\perp, z) = u_{2D}(x_\perp - h(z)),$$

where u_{2D} is the straight tube’s planar field: each z -slice is the same solenoidal planar field, merely translated. This ansatz has zero divergence, and its curl is the tilted core to $O((\kappa_j d)^2)$ — but Biot–Savart is determined by the curl, so the companion’s actual field differs from the ansatz only at $O((\kappa_j d)^2) \cdot \Gamma_j/d$. The ansatz has $u_z \equiv 0$, hence its axial strain component vanishes *identically*: $S_{33} = \partial_z u_z \equiv 0$ (machine-certified in matrix form, `cp_veto_shifted_field`). Its off-axis entries S_{13}, S_{23} are first order in the tilt h' , of size $O(\kappa_j d \cdot \Gamma_j/d^2)$; and in the pairing of tube i , whose axis makes angle θ_{ij} with the companion’s, the expansion of $\xi^\top S \xi$ carries *no* $\cos^2 \theta$ term when $S_{33} = 0$ (machine-certified, `cp_axial_residue`): every surviving term is at least first order in $\sin \theta_{ij}$ against entries themselves first order in tilt, or second order outright. Collecting orders,

$$\alpha_{ij} \leq C_v (\theta_{ij} + \kappa_j d)^2 \frac{\Gamma_j}{d^2},$$

for parallel and anti-parallel geometry alike (θ is taken mod π), with C_v from the kernel constants of Section 13.2. $\square$ (i)

(ii) *The equilibrium payment.* Call the pair *mutually dominated* when $\Gamma_j/d^2 \geq \gamma$, i.e. $d \leq d_{\gamma,j} = (\Gamma_j/\gamma)^{1/2}$. There the mutual strain is the organizing strain, with gap $g_{\text{pair}} = c_g \Gamma_j/d^2$, and Section 14.2 pins a persistent pair at

$$\theta_{\text{eq}} \leq C \frac{\gamma}{\Gamma_j/d^2} = C \frac{\gamma d^2}{\Gamma_j}$$

— the equilibrium tightens with the pair’s own intensity. By (i) at equilibrium, the angular part of the payment is

$$C_v \theta_{\text{eq}}^2 \frac{\Gamma_j}{d^2} = C_v C^2 \frac{\gamma^2 d^2}{\Gamma_j} \leq C_v C^2 \gamma \quad (d \leq d_{\gamma,j}),$$

(the evaluation at the dominance edge is machine-certified, `cp_payment_algebra`). Summing over the band with the circuit-stack count: a WINDING class circuit is a stack of $\geq L_j/(4\delta_j)$ disjoint core balls, each carrying the circulation floor $\Gamma_j^2/(\pi\delta_j)$ (Section 15.3’s currency; the

stack DERIVES $c_E = 1/4\pi$), so $E_j \geq \Gamma_j^2 L_j / (4\pi \delta_j^2) \geq \Gamma_{\min}^2 / (2\delta_c^2)$ using the winding floor $L_j \geq 2\pi$ (non-contractible on $\mathbb{T}^3$) and the core ceiling $\delta_c^2 \leq 4\nu / (c_1 \gamma)$ from the strain band of (C2) (a tube below the band fattens past δ_c , falls below ω^* , and is culled — clause (C5)); hence

$$N_{\text{band}}^W \leq \frac{4\Omega \delta_c^2}{\Gamma_{\min}^2} \leq \frac{16\nu\Omega}{c_1 \gamma \Gamma_{\min}^2}, \quad \sum_{\text{band}} d_{ij}^2 \leq N_{\text{band}}^W \cdot \frac{\Gamma_{\max}}{\gamma},$$

and the band total is

$$\sum_{\text{band}} \alpha_{ij} \leq C_v C^2 \frac{\gamma^2}{\Gamma_{\min}} \cdot N_{\text{band}}^W \cdot \frac{\Gamma_{\max}}{\gamma} = C_n^W \nu \Omega(t), \quad C_n^W = \frac{16 C_v C^2 \Gamma_{\max}}{c_1 \Gamma_{\min}^3} :$$

the γ 's cancel identically (machine-certified, `cp_band_cancellation`). The bend part of (i) contributes $C_v(\kappa_j d)^2 \Gamma_j / d^2 = C_v \kappa_j^2 \Gamma_j \leq c \omega_j / M^2$ by (C1) — a small-class-constant multiple of the core scale, paid into the collapse margin of Section 14.3 exactly as its θ^{*2} term, with the same one-directional choice of M . Pairs in the ambient-dominated band $d \in [d_{\gamma,j}, \ell_N]$ contribute at most γ each by the definition of $d_{\gamma,j}$ and are counted by the same enstrophy weight, landing in the same $C_n^W \nu \Omega(t)$ form. $\square$ (ii)

(iii) *Transients*. A pair not at equilibrium (a crossing) contracts toward it at the mutual organizer's rate $g_{\text{pair}} = c_g \Gamma_j / d^2$ by Section 14.2, while its unvetoes payment is at most $\theta^2 \Gamma_j / d^2$: the integrated cost of one transient is

$$\int \theta(t)^2 \frac{\Gamma_j}{d^2} dt \leq \theta_0^2 \frac{\Gamma_j}{d^2} \cdot \frac{1}{2g_{\text{pair}}} = \frac{\theta_0^2}{2c_g} \leq C_t,$$

an absolute constant — the organizer's own magnitude sets the contraction clock and cancels. Events are finitely many per tube per episode: concurrent band partners number at most the Besicovitch constant b_3 ; re-entry into the band is driven by the ambient strain (slow clock, rate γ , against episode length c/γ : $O(1)$ re-entries); and a merger ends both parents' identities (clause (C3)), so a given identity merges at most once. Total per-episode transient cost $\leq C_{\text{pair}} = b_3 C_{\text{re}} C_t + C_{\text{merge}}$, absorbed multiplicatively in the episode chain: $C'_1 \rightarrow C'_1 + C_{\text{pair}}/c$. *The compact-circuit carriers, by case*: a RING (compact circuit of diameter D) is dipole-suppressed beyond $2D$ (Section 15.3's moment identity and expansion) and self-propels at the classical speed $\sim (\Gamma/4\pi R_{\text{ring}}) \log$, exiting any band shell on its own clock — an event, counted as above. A LOAD-BEARING kink ($\omega \geq R_7 \gamma$) is a bin transient: its residence is at most N^* quiet windows of length $\sim c/\omega$, so its lifetime line-rate integral obeys $\int |S| dt \leq (\Gamma/d^2) \cdot N^* c/\omega \leq [\Gamma_{\max} N^* c C'_1 / (4\nu R_m^2 R_7)] \gamma$ using $d^2 \geq 4\nu R_m^2 / (C'_1 \gamma)$ and the clock ratio $\gamma/\omega \leq 1/R_7$ — γ -multiplicative, absorbed by the R_m -clause. A SUB-LOAD-BEARING carrier is fat — flux forces $s^2 \geq \Gamma/(\pi\omega)$ — and pays $\Gamma/d^2 \leq \pi\omega s^2 / (R_m^2 s^2) \leq (\pi R_7 / R_m^2) \gamma$ when separated ($d \geq R_m s$), absorbed by $R_m^2 \geq 24\pi b_3 R_7$; when not separated, (C3) (extended to all $\Gamma_{\min}$ -carriers) makes the configuration one structure undergoing a merger — an event. $\square$ (iii)

(iv) *The far field, over cells*. Partition $\mathbb{T}^3$ into cells of side $\ell_N = c_N N^{-1/3}$ — the mean spacing from the γ -free carrier count: each carrier crossing its cell pays the circulation floor $\Gamma_{\min}^2 / (\pi \ell_N)$ (Section 15.3's currency), so $N \leq 4\pi \ell_N \Omega / \Gamma_{\min}^2$, and the mean-spacing solve gives $N = (4\pi c_N \Omega / \Gamma_{\min}^2)^{3/4}$ (machine-certified, `r2b_count_solve`); *cells pack by construction, no minimum-separation hypothesis on tubes enters anywhere, and no strain quantity enters the count*. For a cell Q at dyadic shell distance $\sim 2^k \ell_N$ from tube i , the kernel bound of Section 13.2 gives $|S[\omega \mathbf{1}_Q](x_i)| \leq C (2^k \ell_N)^{-5/2} \|\mathbf{u}\|_{L^2(Q)}$; the shell holds at most $c \cdot 2^{3k}$ cells;

Cauchy–Schwarz over the cells of each shell and geometric summation give

$$\gamma_i^{\text{far}} \leq C_\gamma \left(\sum_{k \geq 0} 2^{3k} \cdot (2^k \ell_N)^{-5} \right)^{1/2} \|\mathbf{u}_0\|_{L^2} = C_\gamma \left(\frac{4}{3} \right)^{1/2} \ell_N^{-5/2} \|\mathbf{u}_0\|_{L^2},$$

the series $\sum 4^{-k} = \frac{4}{3}$ machine-certified (`w9_shell_series`) and $\|\mathbf{u}\|_{L^2} \leq \|\mathbf{u}_0\|_{L^2}$ by the energy identity. Sub- ℓ_N vorticity is the near band, already paid by (ii)–(iii) and by Section 15.2. $\square$

15.2. Proof of Lemma 7.2. (i) *The threshold and the log bound.* The threshold $\omega^* = c_* \omega_{\text{rms}}(t)$ is a measurable function of the solution’s present state and never depends on γ : the feared self-reference is severed at the definition. Decompose $\omega = \omega_{\text{class}} + \omega_{\text{cx}} + \omega_b$ with $|\omega_b| \leq \omega^*$; every unit of vorticity lies in exactly one part. The budget consumes *core-averaged* strain, and averaging against the core profile mollifies the Biot–Savart strain kernel at scale δ_i : $|K_{\delta_i}(z)| \leq C \min(\delta_i^{-3}, |z|^{-3})$, with the spherical cancellation retained for $|z| \geq \delta_i$. Split B_{ℓ_N} into dyadic shells from δ_i to ℓ_N ; on each shell Cauchy–Schwarz gives the switch

$$|\langle S[\omega_b \mathbf{1}_{\text{shell}}] \rangle_i| \leq C (2^{-j} \ell_N)^{-3/2} \min(\omega^* \text{vol}^{1/2}, (2\Omega)^{1/2}),$$

and the L^∞ branch pays exactly $C\omega^*$ per shell: summing the $\log_2(\ell_N/\delta_i)$ shells,

$$|\langle S[\omega_b \mathbf{1}_{B_{\ell_N}}] \rangle_i| \leq C_b \omega^* \log(e + \ell_N/\delta_i);$$

the sub- δ_i contribution is one mollified scale ($\leq C\omega^*$, no logarithm), and beyond ℓ_N the background rides inside the cells of Section 15.1(iv). $\square$ (i)

(ii) *The absorption, with structural coefficient.* By the Burgers band, $\delta_i \geq (4\nu/C'_1\gamma)^{1/2}$, so γ enters the bound only inside the logarithm: the budget inequality has the shape $\gamma \leq A + B \log(e + C_\delta \gamma)$ with A, B as in Theorem 7.3. The elementary factorization

$$e + C_\delta \gamma \leq (e + C_\delta(A + B)) \left(1 + \frac{\gamma}{A + B} \right)$$

holds identically (the surplus is $C_\delta(A + B) + e\gamma/(A + B) > 0$). The *calibrated* bound $\log(1+t) \leq \varepsilon t + \log(1/\varepsilon)$ for $0 < \varepsilon \leq 1$ (machine-certified, `bg_calibrated_log`; at $\varepsilon = 1$ it is $\log(1+t) \leq t$) splits the logarithm with the coefficient of γ on the right equal to $B\varepsilon/(A + B)$. At $\varepsilon = 1$ the coefficient is $B/(A + B) < 1$ — *structural, not tuned* — and the absorbed form

$$A\gamma \leq (A + B)(A + B \log(e + C_\delta(A + B)))$$

is machine-certified (`bg_log_absorption`). The budget’s closure (Section 15.4) takes $\varepsilon = \min(1, \frac{3(A+B)}{8B})$: the coefficient is then at most $\frac{3}{8}$ in *every* regime — including $B > A$, where ω^* dominates the small-entropy budget — at calibration cost $\log(1/\varepsilon) \leq \log \frac{8}{3} \leq 1$, because $B < A + B$ and $\frac{8}{3} < e$ (machine-certified, `bg_split_coefficient`). $\square$ (ii)

(iii) *The intense sub-circulation residue.* Structures with $|\omega| > \omega^*$ but $\Gamma < \Gamma_{\min}$ are of two kinds. *Culled remnants* (exits and reconnection debris) are born in events already counted by Section 15.1(iii); they are productionless (Theorem 6.5: no sustained pairing without organization), so they decay on their viscous core time δ^2/ν , and their integrated strain on tube i over that lifetime is at most $C\Gamma_{\min}/(\nu R_m^2)$ per event — an absolute constant, absorbed into C_{pair} . *Wrapped satellites* — vorticity wound azimuthally about the tube, the coaxial-ring geometry whose axial push the parallel veto does not suppress (a ring drives axial flow on its axis; cf. the core-dynamics phenomenology [MH94]) — accumulate at most

a ledger-bounded circulation $\Gamma_{\text{sh}} \leq C_{\text{pair}}\Gamma_{\text{min}}$ per episode (each capture is a counted event, and the sheath turns over viscously within the episode); their axial strain is then

$$\frac{\Gamma_{\text{sh}}}{(R_m\delta_i)^2} \leq \frac{C_{\text{pair}}\Gamma_{\text{min}}C'_1}{4\nu R_m^2} \gamma = \lambda\gamma,$$

using the Burgers relation $\delta_i^{-2} \leq C'_1\gamma/4\nu$ — a *multiplicative* term. The merger-floor constant R_m is free, data-only, and chosen with $\lambda \leq \frac{1}{24}$: the sheath's coefficient joins the calibrated logarithm split (at most $\frac{3}{8}$) in the joint closure of Section 15.4, total at most $\frac{1}{2}$, and the absorption $\gamma \leq X + \frac{1}{2}\gamma \Rightarrow \gamma \leq 2X$ is machine-certified (`bg_half_absorption`). $\square$

15.3. The capped-debris pricing. The residue ω_{cx} 's stretching contribution at the envelope argmax, integrated over capped residence, is bounded by an additive constant.

Near a class tube the local strain is the companion's shifted two-dimensional field with eigenvalues $(\sigma, 0, -\sigma)$ (Section 15.1(i)); the middle eigenvalue is the vetoed axial one, so the cap itself bounds sustained growth by $c_\lambda\gamma + m$ for any orientation of ξ — this branch needs no pricing.

Debris. Every intense non-class identity is born in a counted event (clause (C5); Section 15.1(iii)) with circulation $\Gamma_j < \Gamma_{\text{min}}$ and core δ_j . By Section 14.4 it is productionless while unorganized — sustained pairing without organization is excluded — so it either decays on its viscous core time δ_j^2/ν or re-organizes, a counted transition into the class. Its lifetime strain integral at *any* point is

$$\int |S_j| dt \leq \frac{\Gamma_j}{\max(d, \delta_j)^2} \cdot \frac{\delta_j^2}{\nu} \leq \frac{\Gamma_j}{\nu} \leq \frac{\Gamma_{\text{min}}}{\nu},$$

uniform in the distance d (machine-certified, `cd_event_integral`) — the near-core case saturates at exactly Γ_j/ν — so arbitrary overlap of debris clusters is covered: the pricing is integral, not rate-wise. Events over $[a, b]$ number at most $\sum_{\text{ep}} N_{\text{band}} O(1 + c_r \int_{\text{ep}} \gamma)$, and the episode sum cancels γ exactly as in Section 15.1(ii) (machine-certified, `cd_event_count`), giving a data-only count. Hence $C_{\text{CD}} = (C\Gamma_{\text{min}}/\nu) \cdot N_{\text{events}}(a, b)$, explicit in (E_0, ν) and the interval data. $\square$

15.4. Proof of Theorem 7.3. *The pointwise display.* Collecting: the far field (Section 15.1(iv)) contributes $K'\Omega^{5/6} + K'$ after inserting $\ell_N \geq c'_N(\Gamma_{\text{min}}^2/\Omega)^{1/4}$ (the γ -free count solve, `r2b_count_solve`): the raw far-field power is $\Omega^{5/8}$, dominated by $x^{5/8} \leq x^{5/6} + 1$, so the display keeps its published form with the constant absorbed into K_2 (the historical exponent chain `w10_exponent_chain` remains certified, role demoted); the near band (Section 15.1(ii)–(iii)) contributes $C_n^W \nu \Omega$; the background log bound (Section 15.2(i)) contributes $B \log(e + C_\delta \gamma)$ with $B = C_b \omega^*$; and the multiplicative lines contribute $[\lambda_{\text{sheath}} + \lambda_{\text{carr}} + \lambda_{\text{kink}}]\gamma$, each $\leq \frac{1}{24}$ by the R_m -clauses ($R_m^2 \geq 24\pi b_3 R_7$ -form). The joint absorption — the calibrated split of Section 15.2(ii) at $\varepsilon = \min(1, \frac{3(A+B)}{8B})$, coefficient at most $\frac{3}{8}$, plus the three $\frac{1}{24}$ -lines totalling $\frac{1}{8}$ (machine-certified, `r2asm_allocation`: $\frac{3}{8} + 3 \cdot \frac{1}{24} = \frac{1}{2}$), closed by `bg_half_absorption`, with calibration cost $2B \log \frac{8}{3} \leq 2B$ (machine-certified, `bg_split_closure`) — yields

$$\gamma(t) \leq 2A(t) + 2B(t)[1 + \log(e + C_\delta(A(t) + B(t)))], \quad A = K'\Omega^{5/6} + C_n^W \nu \Omega.$$

Time integration. Hölder in time with the conjugate pair $(\frac{6}{5}, 6)$ — the pairing $\frac{5}{6} + \frac{1}{6} = 1$ is machine-certified (`w10_holder_exponents`) — against the one unconditional integral $\int_0^\infty \Omega dt \leq E_0/2\nu$ gives $\int_0^T K' \Omega^{5/6} dt \leq K T^{1/6}$; the near band integrates directly, $\int_0^T C_n^W \nu \Omega dt \leq C_n^W E_0/2 =: K_2$; and the background term obeys the pointwise bound $\omega^* \log(e + \cdot) \leq 2\Omega^{5/6} + 3$ in class units (its $\Omega^{5/6}$ part joins K , its constant part integrates to $K_3 T$). Hence

$$\int_0^T \gamma dt \leq K T^{1/6} + K_2 + K_3 T \quad \text{for every } T < \infty,$$

with K, K_2, K_3 explicit in (E_0, ν) and class constants.

The drift lemma. Within an episode the budget's sources drift by bounded factors: production is *pairing-capped* — $d\Omega/dt = P - D$ with P concentrated on class cores where $\alpha \leq C'_1 \gamma$ (Section 13.11), null on the background (Theorem 6.5), and event-bounded on transients (Section 15.1(iii)) — so $d\Omega/dt \leq 2C'_1 \gamma \Omega + \text{event terms}$ and over the episode length c/γ every budgeted quantity moves by at most $e^{C_d c}$; hence $\gamma(t) \leq e_d \gamma(t_n)$ throughout, absorbed multiplicatively into the episode constant.

Interval form. Chaining episodes under the organization hypothesis: production is pairing-capped on class cores, margin-paced on capped material, event-bounded on transients, and race-paced on bin occupancies (their enstrophy lags their own peak squared — the quiet-window ratio decay of Section 17.5; entries are event-ledgered), so $d\Omega/dt \leq 2(C'_1 + c_\lambda) \gamma \Omega + 2m \Omega + \text{event terms}$, whence

$$\Omega(t) \leq \Omega(a) \exp\left(2(C'_1 + c_\lambda) \int_a^t \gamma + 2m(t - a) + C_{\text{ev}} N_{\text{ep}}\right) \quad \text{on any organized interval,}$$

with $N_{\text{ep}} \leq 1 + (e_d/c) \int \gamma$ (the exponent assembly: `drift_interval_bound`); the pointwise display then bounds $\sup \gamma$; the event constant C_{ev} enumerates band events, engulfment mergers ((C3) extended), ring transits, grace transits, and bin-occupancy entries — each counted, each ledgered. Nothing in this bound references the vorticity supremum. $\square$

16. COMPLETE PROOFS FOR SECTION 8: THE CHAIN AND THE CRITERION

16.1. Proof of Lemma 8.1. *The forced-damped balance.* The elliptical deformation of a class tube's core is the $m = 2$ mode of the local linear theory of Section 12 about the modulated column. Two rates compete. The ambient strain forces the mode at $\|F_{m=2}\| \leq C_f \gamma$ (the forcing decomposition of Section 13.11, Step 2, restricted to the $m = 2$ sector). The damping is the column's own: purely inviscid mixing of an $m = 2$ disturbance is algebraic [BCV19; Sch+00], but the strained viscous core damps the quasi-mode exponentially: the load-bearing floor is the class constant $c_2 > 0$ of Lemma 4.6 (discrete strained spectrum, no neutral $m = 2$ modes, compactness over the class family), whose sharp large-Reynolds form is the enhanced-dissipation rate $\sim \omega_0 \text{Re}_\Gamma^{-1/3}$ [LWZ20; Gal18], where $\text{Re}_\Gamma = \Gamma/\nu$ is the circulation Reynolds number. For class tubes, (C2) bounds $\text{Re}_\Gamma \in [\Gamma_{\min}/\nu, \Gamma_{\max}/\nu]$ — a *class parameter*, not a growing quantity — so the sharp form gives the floor

$$c_2 \geq c(\nu/\Gamma_{\max})^{1/3}$$

(the antitone bound $x \leq B \Rightarrow x^{-1/3} \geq B^{-1/3}$ is machine-certified, `ec_rate_floor`). The ellipticity therefore obeys

$$\frac{de}{dt} \leq -c_2 \omega_0 e + C_f \gamma, \quad e(t) \leq e(0) e^{-c_2 \omega_0 t} + e_{\text{eq}}, \quad e_{\text{eq}} = \frac{C_e \gamma}{\omega_0}, \quad C_e = \frac{C_f}{c_2},$$

by the explicit solution of the linear comparison equation — the equilibrium ellipticity vanishes linearly in the clock ratio γ/ω_0 , in agreement with the classical strained-vortex equilibria at weak strain [MS71; Kid81].

The perturbative bootstrap. The $m = 2$ linear theory is valid for $e \leq 1/M$. For tubes in the load-bearing regime $\omega_0 \geq R_7\gamma$ with $R_7 = 2MC_e$,

$$e_{\text{eq}} = \frac{C_e\gamma}{\omega_0} \leq \frac{C_e\gamma}{2MC_e\gamma} = \frac{1}{2M}$$

(machine-certified, `ec_threshold`) — strictly inside the validity window — and since the relaxation is a contraction at the fast rate $c_2\omega_0$, a tube entering the regime with $e \leq 1/M$ is pulled to $e \leq 1/(2M)$ within the fast time and held there: the bootstrap closes, and clause (C1)'s near-circularity in the load-bearing regime is *derived*. Below the clock ratio no statement is made — none is needed, by the split of Section 16.3. $\square$

16.2. Proof of Lemma 8.2. *The frame.* By the corrected identity of Section 3.1, at any spatial maximizer of $|\omega(\cdot, t)|$ the transport term vanishes and $\xi \cdot \Delta \omega \leq 0$, so the Danskin–Dini derivative of $M(t) = \|\omega(\cdot, t)\|_\infty$ obeys $D^+M \leq \alpha_{\text{env}}M$ whenever $M > 0$: for $t_n \downarrow t$ realizing the upper limit, maximizers x_n of $|\omega(\cdot, t_n)|$ accumulate at a maximizer $\bar{x}$ of $|\omega(\cdot, t)|$, and $M(t_n) - M(t) \leq |\omega(x_n, t_n)| - |\omega(x_n, t)|$, with the right side governed at x_n by the one-sided form (machine-certified, `l1_subsolution_sign`). The same holds for the interior maximum of any connected component of $\{|\omega(\cdot, t)| > \tilde{r}(t)/c_F\}$: boundary values equal the threshold, so component maxima above it are attained in the interior; births enter at the threshold, merges take the larger parent, splits inherit, and the value is continuous through every surgery. Grönwall in the Dini form integrates the bound.

Regime coverage. Above $\tilde{r}$ every maximizer is intense ($M > \omega^*$) and load-bearing ($M > R_7\gamma$) by construction, hence classified at its active scale (Section 14.1); the time in $[a, b)$ partitions by that classification.

Class residence. $\alpha \leq C'_1\gamma$ pointwise on cores throughout episodes (Section 13.11) — the cap is a statement about the core, not about a path.

Capped residence. At tube-dominated maximizers the middle strain eigenvalue is the vetoed axial one — zero at leading order for two-dimensional strain (machine-certified, `ft_middle_eigenvalue`) — so $\alpha_{\text{env}} \leq c_\lambda\gamma$ plus vetoed corrections; at background-dominated maximizers the background is priced by Section 15.2, mollified at the persistence scale $\delta_x = \sqrt{\nu/M}$ (the scale exists by Section 17.5: heat-surviving structure is $\sqrt{\nu}$ -fat in window units); at debris-dominated maximizers the contribution is integral-priced, $\int \alpha_{\text{cx}} dt \leq C_{\text{CD}}$ (Section 15.3). The within-window margin is $C_d\gamma$, absorbed by the drift mechanism of Section 15.4.

Transits. An occupancy of the organizing horn contracts on the organizer's own clock: duration at most $g^{-1} \log(1/\theta^*)$ at growth rate at most g/c_o , so the per-burst factor is θ^{*-1/c_o} , independent of the intensity and of g (machine-certified, `ft_transit_factor`); gap-degenerate bursts are pancake-type and excluded by the fork of Section 14.3. An occupancy of the arranged-tangled horn lasts at most N^* maintenance windows with total factor F_{tan} (Section 17.5). *Entries:* material entering intensity from below $\tilde{r}/c_F$ during a single occupancy gains at most c_F , so it cannot reach the argmax while $M > \tilde{r}$ — every hosting structure was intense before its occupancy began. *Counting:* under the hypothesis no tangle survives an episode boundary, and every occupancy that resolves at intensity exits as a class tube (Section 14.3); charging each hosted occupancy to its output class identity, occupancies per episode number at most $N_{\text{band}}(1 + c_r \int_{\text{ep}} \gamma)$, and the episode sum pays by the same

enstrophy-weighted γ -cancellation as Section 15.1(ii): at most $C_J + c_r \int_a^b \gamma + N_{\text{ep}}$ in all, with C_J explicit in E_0 and class constants.

The fold. Multiplying the factors along the partition and folding ($e^A G^n = e^{A+n \log G}$, machine-certified, `ft_chain_fold`),

$$M(T) \leq M(0) \exp \left[(C'_1 + c_\lambda + C_d) \int_0^T \gamma dt + mT + C_{\text{CD}} \right. \\ \left. + (\log \theta^{*-1/c_o} + \log F_{\text{tan}}) \left(C_J + c_r \int_0^T \gamma dt + N_{\text{ep}} \right) \right],$$

every γ -integral paid by Section 15.4 and N_{ep} finite on finite intervals. $\square$

16.3. Proof of Theorem 8.4. Let every episode boundary in $[a, b)$ present all intense structures class-organized or capped. The pointwise identity $\max(w - r, 0) + \min(w, r) = w$ (machine-certified, `ec_bkm_split`; valid for any threshold) splits the vorticity supremum at $\tilde{r}(t) = c_F \max(R_7 \gamma(t), \omega^*(t))$:

$$\int_a^b \|\omega\|_\infty dt = \int_a^b \min(\|\omega\|_\infty, \tilde{r}) dt + \int_a^b (\|\omega\|_\infty - \tilde{r})_+ dt.$$

The part below. $\int \min(M, \tilde{r}) \leq c_F [R_7 \int_a^b \gamma + \int_a^b \omega^*]$; the budget pays the first term (Section 15.4), and Cauchy–Schwarz against the energy identity pays the second:

$$\int_a^b \omega^* dt = c_* \left(\frac{2}{(2\pi)^3} \right)^{1/2} \int_a^b \Omega^{1/2} dt \leq c_* \left(\frac{2}{(2\pi)^3} \right)^{1/2} \left((b-a) \frac{E_0}{2\nu} \right)^{1/2} = K_4 (b-a)^{1/2},$$

the constant algebra machine-certified (`k4_omega_star_integral`), *with no appeal to the linear theory*: below the threshold, cores may be as elliptical as they like.

The base. Every maximal super-level interval of M above $\tilde{r}$ begins at $t = a$ or at a crossing where $M \leq \sup_{[a,b]} \tilde{r}$; the interval form of the drift lemma (Section 15.4) bounds $\sup \Omega$, hence $\sup \gamma$ and $\sup \tilde{r}$, explicitly in $(\Omega(a), E_0, \nu, b-a)$ — the drift argument never references the envelope, so no circularity enters. With the range $M \leq M_8$ absorbed, the base is $B_{a,b} = \max(\|\omega(a)\|_\infty, \sup \tilde{r}, M_8)$.

The part above. Every maximizer is intense and load-bearing by construction, provably near-circular in class residence (Section 16.1), and the envelope rides Section 16.2; the organization hypothesis re-supplies Section 13.11 at every boundary, the within-episode drift is absorbed by Section 15.4, and the arranged-tangled occupancies are controlled by Section 17.5. Chaining episodes and collecting the constants of Section 16.2's fold into C_{ch} and K'_2 yields the display of Theorem 8.4, finite for $b < \infty$; hence $\int_a^b \|\omega\|_\infty dt < \infty$, and the Beale–Kato–Majda criterion [BKM84] extends smoothness past b . $\square$

17. COMPLETE PROOFS FOR SECTION 9: THE FUNNEL AND THE SQUEEZE

17.1. Proof of Lemma 9.1. (i) *Well-posedness and the regular horn.* For fixed $t < T^*$ the map $s \mapsto \mathfrak{E}_{x^*}(s, t)$ is continuous (dominated convergence) and vanishes as $s \rightarrow 0$ ($\mathfrak{E} \leq \kappa_3 s^4 M(t)^2$ for smooth fields), so $r_c(t)$ is well defined and positive, and when $r_c(t) < r_0$ the threshold is *realized*: $\mathfrak{E}_{x^*}(s_c(t), t) = \mathfrak{E}^*$ at some $s_c(t) \leq r_c(t)$, by the intermediate value theorem; the cylinder form $\mathfrak{E}_{\text{cyl}}(s, t) := \sup_{t-s^2 \leq \tau \leq t} \mathfrak{E}(s, \tau)$ has the same properties (a sup of a continuous function over a compact, continuously varying interval), and the windows below use it. Suppose now $\liminf_{t \rightarrow T^*} r_c(t) = 2r_\infty > 0$, so $\mathfrak{E}(s, t) \leq \varepsilon_* \nu^2$ for all $s \leq r_\infty$ and t near T^* . Two displayed estimates close regularity. *Riesz localization*: by the

dyadic-shell decomposition of the strain kernel (the Section 15.3 toolbox), $\|S\|_{L^2(B_s(x^*))} \leq C_R \varepsilon_*^{1/2} \nu s^{-1/2}$ plus far-energy terms vanishing with s : the scale-critical strain is $\varepsilon_*^{1/2}$ -small. *Moser absorption*: in the window's Galilean frame (the sweep constant of the harmonic splitting; the vorticity equation is frame-invariant and the drift $|\mathbf{u} - c|$ is locally bounded from $\mathfrak{E}$ -smallness), the first Caccioppoli step on $|\omega|$ with cutoff φ at scale s absorbs its stretching term by Gagliardo–Nirenberg and Young:

$$2 \int \varphi^2 \omega \cdot S \omega \leq \nu \|\nabla(\varphi \omega)\|_{L^2}^2 + C_A (\varepsilon_*)^2 \nu s^{-2} \|\varphi \omega\|_{L^2}^2,$$

so for $\varepsilon_* \leq \varepsilon_*^{\text{reg}}$ the dissipation dominates and the classical parabolic iteration (bounded drift, subcritical forcing) yields $\omega \in L^\infty$ near (x^*, T^*) ; the local Biot–Savart split against the global energy,

$$\|\mathbf{u}\|_{L^\infty(B_\rho)} \leq C(\rho \|\omega\|_{L^\infty(B_{2\rho})} + \rho^{-3/2} (2E_0)^{1/2}) \quad (t\text{-uniform by the energy identity}),$$

bounds $\mathbf{u}$ locally, and Serrin's criterion concludes regularity — contradicting the singularity of x^* . Hence $r_c(t_k) \rightarrow 0$ along boundaries $t_k \uparrow T^*$: concentration is forced.

(ii) *The windows*. Take the realized scales $s_k := s_c(t_k) \leq r_k := r_c(t_k)$ (cylinder form). (I1) *by construction*: $\mathfrak{E}(s, \tau) \leq \mathfrak{E}^*$ for every $s \leq r_k$ throughout the window's time slab. *Class bounds on every Q_R* : scales up to $\sigma(Rs_k)$ split by the cap ratio $\beta_k = r_k/s_k$ — if $\beta_k \rightarrow \infty$ the capped range covers every fixed R ; if β_k stays bounded, the super-cap scales carry $\mathfrak{E}(s) \leq 2s\Omega$ with Ω race-bounded (Section 17.5's bootstrap), which is exactly the jet-order closure of the swept Morrey estimate; in both horns the jet-swept profiles obey uniform Morrey, gradient (the swept local energy inequality), cubic (Caffarelli–Kohn–Nirenberg interpolation), and pressure (local decomposition, harmonic interior estimates) bounds $\mathcal{K}(R, \text{data})$. *The exact floor*: with $\hat{\omega}_k(y) = s_k^2 \omega(x^* + s_k y)$, $\int_{B_1} |\hat{\omega}_k|^2 dy = s_k \int_{B_{s_k}} |\omega|^2 = \mathfrak{E}(s_k, t_k) = \mathfrak{E}^*$ exactly (the rescaling power $r^4 \cdot r^{-3} = r$, machine-certified, **rescaled enstrophy power**). *Jet-invariance*: the jet subtraction changes the profile's vorticity by the jet of $\nabla \times w = m_\sigma - \nabla h$, and $s_k^2 \sup |\nabla \times P_{n,k}| \leq C(s_k/\sigma)^2 (\mathfrak{E}^*)^{1/2} \rightarrow 0$ (the harmonic-splitting bounds): the floor passes to the quotient limit verbatim. $\square$

17.2. Proof of Lemma 9.2. Compactness. The class $\mathcal{K}$ carries, by Lemma 9.1(ii), uniform $L_\tau^\infty L_y^2 \cap L_\tau^2 H_y^1$ bounds on Q_2 (velocity, jet-swept), $\iint |\Phi|^3$ and $\iint |p|^{3/2}$ bounds (interpolation; the local pressure decomposition with the harmonic part controlled by interior estimates against the energy), and $\partial_\tau \Phi$ bounded in H_{loc}^{-2} through the equation. Aubin–Lions gives strong $L^3(Q_1)$ compactness of the velocity component; at the vorticity level the windowed enstrophy identity's bounds give $L_\tau^\infty L^1 \cap L_\tau^2$ -control of $\nabla \times \Phi$ with ∂_τ again equation-bounded, so the curl component of $\tilde{X}$ is compact as well. Suitability passes to strong L^3 limits [Lin98; KS07], and the exact floor passes by the strong $L^2(B_1)$ convergence of the curl. Hence $\mathcal{K}$ is compact in $\tilde{X}$, and its covering number $N(\varepsilon)$ is finite; it is *data-only* because $\mathcal{K}$ is cut out by data-only bounds and the equation — no solution-dependent quantity enters its definition.

Recurrence. The window profiles lie in $\mathcal{K}$ up to the $o(1)$ jet-curl correction of Section 17.1(ii), so for k large they lie within $\varepsilon/2$ of $\mathcal{K}$'s $\varepsilon/2$ -net: among any $N(\varepsilon/2) + 1$ windows, two profiles are ε -close in $\tilde{X}$; and since there are infinitely many windows and finitely many net balls, some ball contains infinitely many — along that subsequence the profiles recur to a fixed $\Phi_0 \in \mathcal{K}$. $\square$

17.3. Proof of Lemma 9.3. *The two branches.* Along the recurring subsequence the consecutive selected windows are related by the parabolic rescaling with ratio $\lambda_j = s_{k_{j+1}}/s_{k_j} \in (0, 1)$: the $(j+1)$ -th profile is the λ_j -rescaling of the j -th window's data at the later boundary. If a sub-subsequence has $\lambda_j \rightarrow \lambda \in (0, 1)$, pass to it: both profiles converge to Φ_0 in $\tilde{X}$ while the rescaling map R_λ is continuous on the compact $\mathcal{K}$, so $\Phi_0 = R_\lambda \Phi_0$: the limit is discretely self-similar with factor λ . If instead $\lambda_j \rightarrow 1$, finite products of the ratios realize every factor in a neighborhood of 1 along the diagonal, and the same identification gives $\Phi_0 = R_\rho \Phi_0$ for all ρ near 1: the generated scaling group is continuous, and the limit is exactly self-similar.

Smoothness and smallness. Off-center control is inherited: $B_s(y) \subset B_{s+|y|}(x^*)$ gives $\mathfrak{E}_y(s) \leq \mathfrak{E}^*$ at every point of every fixed ball, so the regular-horn machinery of Section 17.1(i) applies to the limit itself: Φ_0 is smooth with $|\nabla \times \Phi_0| \leq C\varepsilon_*^{1/2}\nu$ at unit normalization on compacts. Ancientness, suitability, and the local-energy class come from the extraction; the exact floor from Lemma 9.2. $\square$

17.4. Proof of Theorem 9.4. *The checklist.* The limit Φ_0 of Lemma 9.3 is: ancient; suitable; in the local-energy class with uniform bounds; *smooth*, with uniformly small vorticity; and exactly self-similar (continuous branch) or discretely self-similar with factor λ (discrete branch). These are precisely the hypotheses of the cited rigidity theorems — here they are *derived*, not assumed.

The contradiction. In the continuous branch, the rigidity of Nečas–Růžička–Šverák and Tsai [NRŠ96; Tsa98] gives $\Phi_0 \equiv 0$. In the discrete branch, the discretely-self-similar rigidity [Tsa14; CW17] gives the same. Both contradict the exact floor $\int_{B_1} |\nabla \times \Phi_0|^2 = \mathfrak{E}^* > 0$ (Lemma 9.2). Hence no recurrent concentration limit exists; and by Lemma 9.1(i), the only alternative at x^* is regularity. No singular point exists. (*The previous version's defect-carrying quantitative squeeze is superseded by this exact endgame; its certified algebraic cores remain in the certificate tree with their roles noted.*) $\square$

17.5. Proof of Theorem 8.3. *Windows and the Duhamel split.* Fix a time t at which the envelope $\arg\max$ is intense, set $r = M(t)^{-1/2}$, and rescale: $\hat{\omega}(y, s) = r^2 \omega(\bar{x} + ry, t + r^2 s)$ on $B_R \times [-1, 0]$, so $|\hat{\omega}(0, 0)| = 1$ exactly and ν is invariant. By the corrected identity of Section 3.1, $|\hat{\omega}|$ is a sub-solution: $\partial_s |\hat{\omega}| + \hat{V} \cdot \nabla |\hat{\omega}| \leq \nu \Delta |\hat{\omega}| + \hat{\alpha} |\hat{\omega}|$ — pressure-free, so the estimate class is lighter than the velocity apparatus of this section. Duhamel against the drift-adjusted heat kernel (the advection handled by the swept local-energy treatment on Lemma 9.2's class, as in Section 17.1(ii)) splits $1 = |\hat{\omega}(0, 0)| \leq H_\sigma + P_\sigma$ into a heat part and a production part, with two consequences: *sup-contraction*, $H_\sigma \leq \sup |\hat{\omega}(\cdot, -\sigma)|$ (heat and transport preserve suprema), and the $L^1 \rightarrow L^\infty$ bound $H_\sigma \leq C(\nu\sigma)^{-3/2} \int_{B_R} |\hat{\omega}(\cdot, -\sigma)| + \text{tails}$.

No productive windows above M_8 . The rescaled local enstrophy vanishes at high intensity:

$$\int_{B_\rho} |\hat{\omega}(y, s)|^2 dy = r \int_{B_{\rho r}} |\omega|^2 dx \leq \frac{2\Omega_{\text{sup}}}{\sqrt{M}},$$

the exponent arithmetic being $r^4 \cdot r^{-3} = r$ (machine-certified, `rescaled_enstrophy_power`), with Ω_{sup} the interval drift bound of Section 15.4 — no circularity. The kernel-weighted strain is then controlled through $\|\Gamma_{\nu\sigma}\|_{L^2} \propto (\nu\sigma)^{-3/4}$ and $\int_0^1 \sigma^{-3/4} d\sigma = 4$ (machine-certified, `kernel_time_integral`), with $\|\hat{S}\|_{L^2(\text{loc})} \lesssim \|\hat{\omega}\|_{L^2(\text{loc})}$ plus a far-field tail that rescales to $o(1)$ (Section 13.2); no pointwise strain bound is used anywhere. Hence

$$P_1 \leq C_8 \Omega_{\text{sup}}^{1/2} \nu^{-3/4} M^{-1/4},$$

for $M \geq M_8 := 16 C_8^4 \Omega_{\text{sup}}^2 / \nu^3$ every window is *quiet*, $P_1 \leq \frac{1}{2}$ (machine-certified, `m8.threshold_arithmetic` with `quarter_root_bound`). From $1 \leq 1/G_w + P_1$ the per-window growth obeys $G_w \leq 1/(1 - P_1) \leq 1 + 2P_1$ (machine-certified, `quiet_growth_bound`). Below M_8 no control is needed: that range is absorbed into the base of Theorem 8.4.

The two floors. In a quiet window the $L^1 \rightarrow L^\infty$ bound gives the *viscous persistence floor* $\int_{B_R} |\hat{\omega}(\cdot, -\frac{1}{2})| \geq c \nu^{3/2}$: heat-surviving mass at unit rescaled time occupies scale $\sqrt{\nu}$ — spikes die. In a productive window (possible only below M_8) the production integral forces $\|\nabla \hat{V}\|_{L^2} \geq c'$ by the enstrophy–production inequality. Either way the window profile carries a floor $c(\nu) > 0$ in the energy norm of this section — the exact quantity the squeeze bounds — and the floor registers in the metric X of the maintenance hypothesis.

Recurrence by pigeonhole. The window profiles of a sustained-intensity occupancy lie in the compact, equation-constrained class $\mathcal{K}$ of Section 17.2 (the concentration windows of Lemma 9.1, jet-swept); with $N(\varepsilon)$ its ε -covering number (data-only), among any $N(\varepsilon) + 1$ consecutive windows two profiles lie within ε in $\tilde{X}$: the recurrence holds along a subsequence with data-only onset.

The contradiction and the assembly. Suppose an arranged-tangled occupancy hosts the argmax at sustained intensity for more than $N^* := N(\varepsilon_8)$ windows. The pigeonhole supplies the recurrence with a profile carrying the persistence floor $c(\nu)$; the extraction, compactness, and self-similar identification of Sections 17.1 to 17.3 consume only the class bounds, suitability, and the recurrence — the $c(\nu)$ floor plays the exact floor’s role here — so the endgame’s quantitative side yields $\|V\|_{E(B_2)} \leq C' \varepsilon^{1/4}$. Define $\varepsilon_8 := (c(\nu)/C')^4$; for $\varepsilon < \min(\varepsilon_0, \varepsilon_8)$ this contradicts the floor. Hence every such occupancy lasts at most N^* windows, and its total growth factor is $\prod (1 + 2P_1) \leq e^{2N^* \bar{P}} \leq e^{N^*} =: F_{\text{tan}}$ — a class constant (the fold is machine-certified, `ftan_fold`; the floor arithmetic, `nu32.floor_arithmetic`; the squeeze arithmetic, `eps8.squeeze_arithmetic`). In particular persistent arranged-tangled occupancy at sustained intensity is impossible at *any* point; the singular-point statement of Theorem 9.4 is the special case in which both floors are available. $\square$

APPENDIX A. THE MASTER CONSTANTS TABLE

Table 2 extends Table 1 with the constraints. Reading: each tier depends only on tiers above it and the data (E_0, ν) ; the graph is acyclic; every “large”/“small” constraint is one-directional; solution-dependent quantities appear only in estimates, never in the choice of a constant.

tier	constants	constraints and provenance
0	$\varepsilon_{\text{CKN}}, b_3, \text{Riesz}$	universal: the CKN ε -regularity constant [CKN82; Lin98]; the Besicovitch overlap [Mat95]; the Hilbert-transform bound
1	E_0, ν	the data
2	$\mathfrak{M}(r_0) \leq 2E_0 + \nu^{-1}E_0; \Gamma_{\max}(E_0)$	the energy identity (M_0 and $\alpha_{\max}$ are retired with the superseded maintenance interpolation: the funnel's class bounds are now supplied by Lemma 9.1(ii) directly)
3	$\Delta_0 \leq \frac{1}{6}; \mu_0; \lambda_0 = \frac{1}{2}; T = 2\log(1/\lambda_0); c_*, c_1, c_E, c_\lambda, c_o, c_r, C_f$	fixed by the linear theory and the definitions; $\Delta_0 \leq \frac{1}{6}$ and the μ_0 -clause are the inheritance conditions of Lemma 6.1
4	$c_2 \geq c(\nu/\Gamma_{\max})^{1/3}; C_e = C_f/c_2; c_f$ (the $m = 1$ spectral floor, Lemmas 4.1 and 4.3); C_{su}	derived: the bounded circulation Reynolds number (Lemma 8.1); the $m = 1$ spectral floor (Lemmas 4.1 and 4.3); sector uniformity (Lemma 4.4)
5	M large; θ^* small	free, one-directional: the perturbative windows of Sections 4 and 5
6	$R_7 = 2MC_e; \Gamma_{\min} = (4c_{\text{cc}}R_7/C_1')\nu$ (Section 14.3); M_8, M_9 (the quiet-window and transport thresholds); $K(\Delta_0, \mu_0), \theta = [\Delta_0\mu_0/(2C_h)]^2, \eta' = -\log_2(1 - \varepsilon_r), n(\eta)$ (the gradient-dichotomy, race, and jet-order constants); $N^* = N(\varepsilon_8)$ (the covering number of Lemma 9.2's class); C_{av} (Section 14.3); F_{tan}	large/derived: the circularization bootstrap (Lemma 8.1); subsumes the collapse and transient-dominance thresholds
7	R_m large ($\lambda \leq \frac{1}{24}$ in Lemma 7.2, with the carrier and kink clauses $R_m^2 \geq 24\pi b_3 R_7$); R_v	large: the merger floor; the sheath absorption
8	m small: $\varepsilon(m) < \min(\varepsilon_0, \varepsilon_8); \varepsilon_* \leq \varepsilon_*^{\text{reg}}$ (the funnel threshold $\mathfrak{E}^* = \varepsilon_*\nu^2$; ε_7 retired with the quantitative squeeze)	small: the squeeze (Theorem 9.4); no lower bound on ε exists anywhere in the argument

TABLE 2. The dependency order of the named constants, with constraints.

Two gauges organize the $\mathfrak{E}$ -denominated constants: the *regularity gauge* $\varepsilon_*\nu^2$ (the funnel: the dichotomy, the Moser horn, the windows, the compact class) and the *ledger gauge* $(2/\pi)\Gamma_{\min}^2$ (the counts, the band, the exports); the filtering theorem of Section 9 partitions the flux range between them at $\Gamma_{\mathfrak{E}} = (\pi\mathfrak{E}^*/2)^{1/2}$.

APPENDIX B. THE CLAIM MATRIX

The development was tracked against a claim matrix: one row per component, each row closed only when its statement reached referee-grade prose with its exact core certified. The matrix below is the closed state; L = Lean certificate (Appendix C), S = symbolic adjudication log, I = calibrated instrument.

component	status	certificates
financing identity (with sign)	exact	L×4, S, I
pancake discriminant identity	exact	L, I
veto; parity kernel	exact	L×2, I
Rayleigh-monotone stability	full	L (core)
2D band: monotone reduction, annulus, depletion, edge	full	I series
viscous band (fixed-dissipation reduction; matched layer and parametrix as the sharp mechanism)	full	L, S, I (rate 0.387)
$m = 1$ sector (translation pair, impulse adjoint)	full	L (Gram), I
sector uniformity; quantified displays	full	L×3, S
organization–production (pairing; null background)	full/exact	L×3, I
alignment rigidity; veto–parity collapse	full	I (basin)
budget identity; moment law; calibration constant	exact	L×5, S
episode estimates (kernel, parity, gain, modulation, adiabatic, bilinear)	full	L×8, S
episode cap $\alpha \leq C'_1\gamma$; smoothing; local existence	full	L ($\beta = \pi$), S
covering and counting; the cell far field	full	L (series), S
close pairs (vetoed kernel, equilibrium payment, transients)	full	L×4, S
background strain (threshold, log bound, calibrated absorption, sheath)	full	L×5, S
the budget $\int \gamma \leq KT^{1/6} + K_2 + K_3T$	full	L, S
circularization; the clock-ratio scoping	full	L×3, S, I
envelope chain (transit and tangled factors, fold)	full	L×5, S
tangled-horn control (M_8 , floors, recurrence, ε_8)	full	L×6, S
capped-debris pricing (per-event integral, event count)	full	L×2, S
classification at the active scale (exterior strain, inheritance)	full	L×2, S
interval regularity; the BKM split	full	L, S
concentration dichotomy and windows (regular horn, exact floor)	full	L×2, S
the compact class and the recurrence	full	S
the self-similar limit (scale-ratio branches)	full	L×2, S
the rigidity endgame (exact rigidity vs the $\mathfrak{E}^*$ -floor)	full	L, S
the main assembly (Section 10)	full	—

APPENDIX C. INDEX OF MACHINE CERTIFICATES

The Lean project contains 89 kernel-checked theorems across nine files, zero unproven placeholders, over a pinned unmodified Mathlib; the build is reproduced by `lake exe cache get && lake build`. Abbreviated index (the repository’s master index carries each theorem’s adjudication log):

theorems	certify
trace_cube_split, trace_SWW, trace_cube_det, financing_pointwise lemma18_pairing, lemma18_cauchy_schwarz, lemma18_org_bound calibration_factor_closed_form qt1_scaling_coefficient, qt1_energy_identity moment_law_assembly, moment_law_equilibrium pancake_extremal_sq, veto_algebra, parity_kernel_odd, rayleigh_core w5_boundary_exponent, w5_window_slack, w6_closure, w3_lognorm_assembly w9_shell_series, w10_holder_exponents, w10_exponent_chain, w9_chebyshev w13_defect_gap w7_duhamel_integral, w1_tail_integral, w8_tail_integral, gram_integral skew_core, hermitian_quadratic_real beta_cont, beta_deriv, beta_integrable, w16_beta_integral sc7_small_eps_choice cp_veto_shifted_field, cp_axial_residue, cp_payment_algebra, cp_band_cancellation bg_log_absorption, bg_half_absorption ec_threshold, ec_bkm_split, ec_rate_floor ft_transit_factor, ft_chain_fold, ft_middle_eigenvalue dc_coherence_inherit sc8_slice_scaling, sc8_holder_exponents su_green_integral, qd_exponent_positive, qd_rpow_collapse l1_subsolution_sign, kernel_time_integral, rescaled_enstrophy_power m8_threshold_arithmetic, quarter_root_bound, quiet_growth_bound	the financing identity: the trace split; the sign $\text{tr}(SW^2) = +\frac{1}{4}\omega^\top S\omega$; $\text{tr } S^3 = 3 \det S$; the assembly the exact pairing $\alpha = \sum A_i \lambda_i$; its Cauchy-Schwarz step; the organization bound $\sqrt{2/3}$ the core-radius ruler factor $\frac{1}{2} - \frac{\ln 2}{3}$ exactly the localized defect energy identity and its exact $-\frac{1}{4}$ scaling balance the second-moment law $4\nu - \alpha m_2$ and the Burgers equilibrium $m_2 = 4\nu/\alpha$ the discriminant inequality with pancake equality; the veto $S\hat{z} = 0$; the odd parity kernel; the Rayleigh sign core the adiabatic boundary exponent $\frac{1}{2}$; the window slack; the bootstrap root bound; the episode-rate assembly the cell-shell series $\frac{4}{3}$; the Hölder pairing $(\frac{5}{6}, \frac{1}{6})$; the exponent chain; the volume Chebyshev bound the dichotomy algebra of the superseded defect-gap argument (statement true; role demoted — see the record) $\int_0^\infty e^{-as} = 1/a$; the kernel tails $r^{-3}/3$, $r^{-5}/5$; the Gram pairing $\langle r, W' \rangle = -2$ the $\mathbb{C}^2$-skew content the Fujita-Kato beta integral $= \pi$, with its three supporting theorems the squeeze arithmetic: $\varepsilon < (\varepsilon_{\text{CKN}}/C_{\text{QT}})^{4/3} \Rightarrow C_{\text{QT}}\varepsilon^{3/4} < \varepsilon_{\text{CKN}}$ the shifted-field veto $S_{33} = 0$; the no-$\cos^2 \theta$ expansion; the equilibrium payment; the γ-free band sum the logarithmic absorption with structural coefficient $B/(A+B) < 1$; the sheath absorption the circularization bootstrap window; the exact split $(w-r)_+ + \min(w, r) = w$; the damping-rate floor the intensity-free transit factor θ^{*-1/c_0}; the chain fold; the vetoed middle eigenvalue the downward coherence inheritance at $3\Delta_0$ the critical-slice scaling $r^2 \cdot r^{-3} = r^{-1}$; the L^2-L^3 Hölder pairing the sector-uniformity kernel integral $1/(m+1)$; the depletion exponent $m - \frac{1}{2} \geq \frac{1}{2}$; its power collapse the v002 envelope layer: the one-sided viscous sign at maximizers; $\int_0^1 \sigma^{-3/4} d\sigma = 4$; the rescaled-enstrophy power $r^4 r^{-3} = r$ the quiet-window theorem: $P_1^4 \leq (1/2)^4$ at $M \geq M_8$; the fourth-root extraction; $G_w \leq 1/(1 - P_1) \leq 1 + 2P_1$

theorems	certify
ftan_fold, nu32_floor_arithmetic, eps8_squeeze_arithmetic k4_omega_star_integral, drift_interval_bound, cd_event_integral, cd_event_count bg_calibrated_log, bg_split_coefficient, bg_split_closure r1a_mean_value_eight, r1b_closure_exponent, r1b_n_eta	the tangled fold $(1+x)^N \leq e^{Nx}$; the persistence-floor mass bound; the ε_8 squeeze arithmetic $C\varepsilon^{1/4} < c(\nu)$ the K_4 constant algebra; the interval-drift exponent assembly; the debris per-event integral $\leq \Gamma_{\min}/\nu$ uniform in distance; the γ -canceling event count the calibrated absorption split: $\log(1+t) \leq \varepsilon t + \log(1/\varepsilon)$; the $\frac{3}{8}$ coefficient pin with cost $\log \frac{8}{3} \leq 1$ ($\frac{8}{3} < e$); the joint closure to the budget display, unconditional the closure wave's kinematic layer: the mean-value chain $\kappa_3 \cdot 6/\pi = 8$; the jet closure exponent $1 + 1/(2n+4)$; the $n(\eta)$ implication
rd_race_ratio, rd_affine_step, rd_rent_window, rd_leakage_clause r2a_floor_sharp, r2b_count_solve, r3a_gamma_min, r3a_ce_stack r3b_rm_clause, r3b_fatness, r4a_overtone, r4c_zero_mean r2asm_allocation, s1_inheritance, pow_dominate, r6_skew_divergence	the race ratio $(1-\varepsilon)^W$; the exact affine fold; rent $\times$ window = $\theta\Omega$; the leakage clause the circulation floor's sharpness; the γ -free count solve; the $\Gamma_{\min}$ cancellation; the $c_E = 1/4\pi$ stack the R_m clauses; flux-forced fatness; the γ -free overtone equilibrium; the zero-mean oscillation identity the $\frac{3}{8} + 3 \cdot \frac{1}{24} = \frac{1}{2}$ allocation; the exterior-strain inheritance arithmetic; the far-field power domination $x^{5/8} \leq x^{5/6} + 1$; the fixed-dissipation skew cancellation

Declared formalization targets (analysis-level; each proved in the analytic text): the torus integration by parts $\int \text{tr}(\nabla \mathbf{u})^3 = 0$; the moment-law and circularization ordinary-differential comparisons; interval-arithmetic eigenvalue enclosures for the swept normal form.

DECLARATIONS

Acknowledgment. The omission of the viscous term in the v001 form of the Section 3.1 display, and the resulting gap in the chain section, were identified by Ember Arlynx, who also contributed Lean formalizations of the maximum-envelope machinery used in the repair (repository pull request #1).

Version note. v002 corrects the kinematic identity of Section 3.1; replaces the per-trajectory chain with the envelope chain (Lemma 8.2 and Theorems 8.3 and 8.4); strengthens the funnel to an any-point theorem with the viscous persistence floor and the threshold ε_8 ; adds the capped-debris pricing (Section 15.3) and the interval drift bound; pins the trailing-window length and the recurrence metric; adds the K_4 term and the corrected display base; states the velocity-only form of the nontriviality floor; completes the budget absorption by the calibrated split of Section 15.2(ii) (unconditional in every regime, the merger floor re-pinned to $\lambda \leq \frac{1}{24}$ in the final allocation); aligns the far-field shell display of Section 15.1(iv) with the exterior kernel bound, making the certified series an exact equality; reissues the classification on exterior strain with the organizer floor (validated on the Burgers exemplar) and the collapse display in window-averaged form; re-architects the funnel on the concentration dichotomy, the jet-swept compact class, and the discretely-self-similar rigidity endgame; derives $\Gamma_{\min}$, $c_E = 1/4\pi$, and the winding-class band constant on the flux-circuit ledger; anchors the linear theory's load on the fixed-dissipation reduction (Lemma 4.6), the $\nu \rightarrow 0$ sharp theory retained as mechanism and instruments; and retires an unused constant from the master table.

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Conflict of interest. The author states that there is no conflict of interest.

Data availability. The Lean formalization (source and build files), the symbolic-adjudication scripts with their logs, and the instrument code with its archived outputs are available in the repository accompanying the paper, <https://github.com/truthintime/navier-stokes-proof>, public upon publication. The complete development record is available from the author upon request.

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